Everyday Social Interactions and Momentary Affective Experiences in Late-Middle Aged and Older Chinese: Using Ecological Momentary Assessment (EMA) to Distinguish Between-Person and Within-Person Relationships

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by

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Abstract of thesis entitled

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Background: Interpersonal environment plays an important role in shaping affective experience during the late adulthood. Most of prior research has focused on how the differences in individuals' overall social relations contribute to affective well-being on a global time-frame. However, it remains unclear that whether and how, within individuals, specific features of social interactions are related to momentary affective experiences on short time-scales.

Objectives: The first primary objective of this study was to develop a mobile-based ecological momentary assessment (EMA) protocol to collect data of social and affective experiences that naturally occur to older people in daily life. The second primary objective was to use the developed EMA protocol to address research questions regarding the between- and within-person relationships of the occurrence and appraisal of social interactions with momentary affect.

Methods: The design and implementation of a mobile-based EMA research protocol was presented, followed by an evaluation of this protocol's feasibility and usability among older people. Using the developed protocol, 78 community-dwelling older adults provided up to seven EMA surveys per day to report on the occurrence of social interactions, their satisfaction with interaction, and positive and negative affect during a one-week period. Multilevel modelling was used for testing the between- and within-person covariation of social interactions and affect and the lagged relations between social interactions and affect.

Results: Participants reported a total number of 3,822 observations (a compliance rate of 91.5% of all assessments), and rated their experiences working on this protocol as relatively pleasant. On a between-person basis, people who had more frequent and more satisfactory interactions also reported a higher average level of HPA. At the within-person level, people experienced greater high- and low-arousal positive affect (HPA and LPA) and fewer high- and low-arousal negative affect (HNA and LNA) when they perceived a social interaction as more satisfactory than usual. Having social interactions at the prior occasion predicted greater HPA but fewer LPA two hours later. Lower interaction satisfaction at the prior occasion predicted greater HNA and LNA two hours later, and the experience of increased HNA and LNA subsequently predicted lower interaction satisfaction at the following interaction.

Discussion: Our study extends existing research by separating the between- and within-person relationships theorized to be occurring in momentary social and affective experiences. On a between-person basis, the positive effect of social interactions was reflected by its connection with a higher average level of HPA. At the within-person level, both the occurrence of and the satisfaction with social interaction has been linked with more favorable affective responses. Interestingly, a lack of social interaction was not necessarily harmful because it played a role in promoting LPA. Furthermore, the mutual reinforcement of negative affect and lower interaction satisfaction was significant, suggesting the reciprocal pattern of negative appraisals of social environment and negative affective experiences. Our findings may direct clinical attentions to the dynamic interplay of social and affective experiences and inform the development of programs that aimed to promote healthier social environments for older adults' maintenance of affective well-being. (498 words)

Keywords: social relations; affective states; real-time ecological assessment; intra-individual variability; older adults;

Declaration

I declare that this thesis represents my own work, except where due acknowledgement is made, and that it has not been previously included in a thesis, dissertation or report submitted to this University or to any other institution for a degree, diploma or other qualification.

Signature ……………………………………..

LIU, Huiying

Date ……………………………………..

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Chapter 1 Introduction

1.1 Background

1.1.1 Maintaining affective well-being as a key component of healthy aging

The substantial growth of aging population is occurring worldwide. According to the latest report released by World Health Organization (2015b), the world's older population (aged 60 years or above) will increase from 900 million in 2015 to reach a total of 2 billion by 2050. While most countries are experiencing a demographic redistribution toward older adults, the pace of population aging in China is even much faster than that of many other countries (Smith, Strauss, & Zhao, 2014). In 2014, the proportion of older population aged 60 years or above was 15.5% in the country, and this number is anticipated to reach 32.8% by 2050 - comprising over 20% of the world's total older population (United Nations, 2013). The Chinese older population will comprise at least one quarter of the world's total population of elders aged 80 and over by the year of 2050 (World Health Organization, 2015a).

One distinctive characteristic of China's aging population is the remarkable size of the population aged between 50 and 70. This age group has accounted for about one quarter of the total population in China today. It was projected that by the year of 2040, this number will reach over 400 million, with an almost even distribution of gender (211 million for male and 202 million for female). At the same time, Chinese people are living longer. The average life expectancy has climbed from 64 years in 1979 to 75.3 years in 2015, which is expected to be at around 80.5 years by 2050-2055 (Guilmoto & Jones, 2016). Given the magnitude of this population as well as the increased life expectancy, how to add a healthy life to this extended later adulthood has become a public health priority (Baltes & Baltes, 1990; S. T. Charles

& Carstensen, 2010; Diener, Chan, & Well‐Being, 2011). Central to this issue is the maintenance of affective well-being, has been increasingly recognized as an indispensable component of healthy aging that involves involves the optimization of opportunities for functional independence that enables well-being and sustained engagement in society during later life (World Health Organization, 2012; Peel, Bartlett, & McClure, 2004).

In the field of aging research, affective well-being has been typically considered one element of subjective well-being, a broader concept involves an individual's "cognitive and affective evaluation of the life he or she was experiencing" (Diener, Suh, Lucas, & Smith, 1999). While the cognitive element of subjective well-being often involves how one thinks about his or her life (e.g., life satisfaction in global terms), the affective element of subjective well-being refers to one's feelings, emotions and moods: affect is considered positive when one's emotions and moods experienced are pleasant and considered negative when one's emotions, and moods experienced are unpleasant (Diener & Larsen, 1993). Here affect is used as an umbrella term that embodies both emotions and moods, and it is important to note that emotions and moods are distinct phenomena: (1) emotions are usually caused by immediate circumstances while moods does not necessarily need a contextual stimulus; (2) emotions come and go quickly whereas moods can often last for an extended period of time; (3) emotions are often experienced more intensively while moods are more stable; (4) in addition, emotions evoke autonomic arousal and may have behavioral implications (for more terminological discussions, see Diener, Scollon, & Lucas, 2009; Lochner, 2016).

Maintaining a higher level of affective well-being, as defined by more frequent positive affect and less frequent negative affect, has been generally cosndiered desirable because it can benefit individual's health and well-being. To date, there is compelling evidence supporting this health promoting view by demonstrating a powerful influence of affect on our health and well-being outcomes (Waldman-Levi, Erez, & Katz, 2015). A long-standing research has linked negative affective experiences with unhealthy patterns of physiological consequences and individuals' increased susceptibility to illness (DeSteno, Gross, & Kubzansky, 2013; Salovey, Rothman, Detweiler, & Steward, 2000). In parallel, there is a rapid accumulation of empirical findings from experimental and large-scale prospective researches demonstrating that positive affect protect against poor health outcomes (Schwerdtfeger & Gerteis, 2014; Stellar et al., 2015). According to this health promoting view, negative affect is not as desirable as positive affect, but there is no doubt that negative affect is experienced by individuals as a normal part of life and serves adaptive functions in areas of cognition and behaviors (Mayer, Gaschke, Braverman, & Evans, 1992). While these adaptive functions were beyond the scope of this study, our conceptualization of affective well-being adopted the view that valence (from negative to positive) is the building block of emotional life (Barrett, 2006b), in which positive affect and negative affect was regarded as two independent factors being expected to contribute to health and well-being in opposite directions.

1.1.2 The implications of social relationships for affective well-being

Social relationships play a fundamental role in shaping affective experiences since the earlies days of our life (Berkman, Glass, Brissette, & Seeman, 2000; Uchino, 2006; Vogel, Ram, Conroy, Pincus, & Gerstorf, 2017). Evidences are emerging that many aspects of individuals' interpersonal environment (e.g., relationship closeness) are even stronger predictors of affective well-being than some widely documented factors such as subjective health and personality traits (Vogel et al., 2017). A substantial number of prospective researches has successfully linked the quantitative (objective) aspects of social relationships (e.g., network size and composition, contact frequency) with the amount and frequency of positive and negative affective experiences (for a review, see Liu, Xie, & Lou, 2018). In general, individuals who have a larger or more diverse social network and those who have more close social ties often reported better affective well-being outcomes than their counterparts who reported small social networks and less close social ties (Fingerman, 2009; Hsieh & Lee, 2014; Schneider et al., 2013; Vogel et al., 2017).

A growing number of studies demonstrated the significance of the qualitative (subjective) aspects of social relationships (e.g., relationship satisfaction, perceived support or helpfulness) for shaping individual affective outcomes (Amieva et al., 2010; Fiorillo & Sabatini, 2011). Recent studies suggested that the qualitative aspects of social relationships were particularly important for understanding affective experiences within specific relationships (e.g., spouse, family, friend) (Kira S Birditt, Miller, Fingerman, & Lefkowitz, 2009; Willson, Shuey, & Elder Jr, 2003). These relationships can either serve as a source where individuals derive comfort and support or generate interpersonal ambivalence and tensions that may boost unfavorable affective responses. Once these qualitative aspects of interactions were considered, the quantitative aspects of interactions become less influential in explaining affective outcomes (Holt-Lunstad, Uchino, Smith, & Hicks, 2007).

Many theories indicated that social relationships tended to be more influential for the maintenance of later-life affective well-being (Fingerman, Turiano, Davis, & Charles, 2013). With older age, individuals experienced a motivational shift that places more focus on emotional goals and the maintenance of emotional closeness with their existing social ties (S. T. Charles & Carstensen, 2010; English & Carstensen, 2014b; Wethington, Pillemer, & Principi, 2016). In addition to motivational shift, older people also experience age-related changes in their ability to negotiate with their surrounding environment, which may reshape the way they interact with social partners and how the environment influence their affective outcomes (Wright & Patterson, 2006). Although older people as a general group, has demonstrated the age-related enhancement of emotion regulation skills, it is also true that they experience greater difficulty than younger people in recovering from exposure to negative interpersonal events (e.g., a conflict) due to the reduced flexibility of physiological systems (Susan Turk Charles & Carstensen, 2008; Sims, Hogan, & Carstensen, 2015; Wrzus, Luong, Wagner, & Riediger, 2015).

1.1.3 The within-person approach to examining affective experiences emergent from ongoing social interactions

Social relationships "take shape and substance in social interactions", which have been defined as "dynamic, changing sequences of social actions between individuals who attach meaning to a contact" (Heatherton & Walcott, 2009; Merrill, 1965). To date, most of the literature linking features of social interactions with affective experiences has adopted a between-person approach, that is, examining how individual differences in features of social interactions (e.g., whether individuals differ from others in the number or frequency of social interactions) are related to individual differences in affective well-being. In line with the assumption that human beings have an inherent need to be socially connected, this research on between-person differences has documented the positive impact of social interactions on affective outcomes (e.g., people have more frequent and pleasant interactions are happier) and the potential negative implications of lacking social interactions for affective outcomes (e.g., people who on average spent more time alone tended to be more sad and depressed).

A conceptually distinct approach is the within-person examination of how specific features of social interactions are related with positive or negative affective states within the same individuals. While the between-person approach is concerned with the trait-level phenomena (e.g., individual features that do not change in the short term), within-person relationship is concerned about states-like phenomena that imply changes unfolding within individuals over time and situations (e.g., a process that implies some kind of within-person variability in certain variables). For instance, whether a person's affective states might differ or change when he or she experience social interactions differently. This within-person perspective helps to understand the interplay of moment-to-moment variations in social interactions and the short-term changes in affective states across situations. Recent research highlighted the importance of addressing these within-person relations because affect is not a static process that can be easily captured by a single or an aggregated score, but demonstrates significant fluctuations within individuals across days, events and moments. In subtle but consistent ways, these fluctuations in affect could influence and be influenced by each of the social interactions people encounter in their daily life (Hülür, Hoppmann, Ram, & Gerstorf, 2015; Vogel et al., 2017).

There has been an accumulation of studies on the within-person covariation of social interactions and momentary affect (Liu et al., 2018). Many studies have focused on the presence of social partners, suggesting that being with others versus being alone would benefit individuals by enhancing more favorable affective experiences, whereas the magnitude (or sometimes the direction) of such association might vary depending on the uniqueness of the presented people (e.g., relationship type, familiarity, importance) (English & Carstensen, 2014a; Hülür et al., 2015; Riediger & Rauers, 2014). Beyond the structural aspects of social ties, a handful of studies has devoted to examining the contribution of the qualitative characteristics of social interaction (e.g., appraisals of an interaction) to the within-person variations in affective states. Although great variations existed across studies in the assessment of subjective appraisals of social interactions, there is agreement that individuals reported greater positive affect when they reported greater pleasantness during the interaction, or when they perceived more communal behaviors of their partners (Boehringer, Schwabe, & Schachinger, 2010; Le & Agnew, 2003; Mehl, Vazire, Holleran, & Clark, 2010). In contrast, elevated negative affect and arousal were reported by individuals when engaging in interactions characterized by disagreement or rejection, and when they perceived more agentic behaviors of themselves or from others (Luong & Charles, 2014; Windsor, Gerstorf, Pearson, Ryan, & Anstey, 2014; Wrzus & Mehl, 2015).

Despite being informative for understanding the within-person interplay of social functioning and affective experiences, this research on within-person relations was still limited regarding the following aspects. First, many of these studies did not properly disaggregate within-person from between-person effects, thus being unable to clarify the relative importance of the long-term structural properties and the short-term process properties of social relations for shaping affective experiences. Second, little efforts have been devoted to addressing the bidirectionality of the within-person covariation of social interactions and affect. While most of existing studies interpreted their findings to support the view that social interactions elicited changes in affect, it remains unclear whether affective states predicted social interactions, for example, whether positive affect people experienced at prior moment would predict more positive appraisals of subsequent interactions and vice versa (Hahn, Cichy, Small, & Almeida, 2013). Third, few of studies on the within-person covariation of social and affective experiences have been conducted with aging population, and even less have paid attention to older people's subjective appraisal of everyday social interactions.

1.1.4 Using "ecological momentary assessment" (EMA) to understand the link between social interactions and affective experiences

On the methodology side, much of the research concerning social interactions and affective experiences was based either on laboratory experimental design or prospective survey design (Cohen, 2004; Deindl, Brandt, & Hank, 2016; Kawachi & Berkman, 2001).While laboratory-based research typically detect individuals' affective responses under a prescribed situation (e.g., a manipulated social conflict), the lack of ecological validity (also named as external validity) is still one of the biggest concern regarding the interpretation of much of these findings. It is likely that the prescribed social events can hardly represent the real-life situations, and it is also problematic to assume that the short-term affective responses observed from laboratory settings are generalizable to day-to-day affective experiences (Wrzus &

Mehl, 2015). In terms of the survey design, the primary concern is about the heavy reliance on the retrospective self-reports that asked participants to provide a global estimate of their social and affective experiences at a single occasion, thus inevitably involving limitations such as recall bias and memory distortion (Beal, 2015; Fraley & Hudson, 2013).

With these concerns, the development of real-time assessment approach such as "ecological momentary assessment" (EMA) (for the original work coined this term, see Stone & Shiffman, 1994) has received increasingly attention. Instead of relying on experimental observations or retrospective reports, EMA was designed to collect real-time assessments of experiences as they occurred in natural environments, and specific to the context of everyday social interactions and affect, to obtain repeated measures on social experiences and affective states, typically multiple times across occasions throughout a day (for more details, see Moore, Depp, Wetherell, & Lenze, 2016). As discussed by pervious researchers, EMA offers a range of benefits for studying underlying mechanisms linking different aspects of everyday experiences: (a) momentary affect can be assessed simultaneously with actual behaviors, appraisals, and contextual variables; (b) repeated assessments can reveal fluctuations in affect over the course of a day, as well as temporal sequencing implied by social interaction and momentary affect (i.e., positive feeling leads to contact with a friend; reduced distress follows discussion of a problem); (c) ecological assessments in people's natural environment enhance generalizability of the results to real life situations (for reviews, see Beal, 2015; Brose & Ebner-Priemer, 2015).

Furthermore, recent advances in mobile technology have fostered the growth of researches adopting more sophisticated mobile-based EMA techniques, in which participants interact with mobile devices to provide real-time data (Collins, Kashdan, & Gollnisch, 2003; Shiffman, 2000). These mobile-based EMA techniques not only overcome several major limitations inherent in paper-based EMA methods (e.g., inaccurate recording of response time, inability for checking faked compliance), and more importantly, expanded the range of EMA studies through incorporating newer features, such as equipping with accelerometers and Global Positioning Systems (GPS) location capture, allowing audio and video recording, connecting to mobile sensors that enables the continuous monitoring of physiological activity in daily life (Brose & Ebner-Priemer, 2015; Case, Burwick, Volpp, & Patel, 2015; Granholm, Loh, & Swendsen, 2008; Hoppmann & Riediger, 2009).

Although there has been enough evidences advocating the use of EMA to the study of everyday social and affective experiences, only a very few number of studies have targeted older population (Rullier et al., 2014). Unfortunately, few of these studies provided information that are necessary for the success of designing and implementing a mobile-based EMA protocol for the use of older population, for example, how to optimize user-based options, how to enhance older participants' confidence in working on the EMA procedures that involve the use of mobile technology. A review of the literature only identified two exceptional studies that reported the use of smartphone-based EMA with satisfactory compliance in the context of North America (Fritz, Tarraf, Saleh, & Cutchin, 2017; Ramsey, Wetherell, Depp, Dixon, & Lenze, 2016), but we still do not know whether similar EMA approaches were also feasible in other cultural context, and more specifically, whether these approaches were acceptable among older Chinese. It is therefore timely and important to provide reports about how to develop a mobile-based EMA protocol among older Chinese regarding the usability and feasibility of the protocol.

1.2 Research objectives

The current study sought to address the research gaps discussed above by developing a mobile-based EMA protocol to examine the momentary characteristics of social interactions in relation to within-person fluctuations in affective states that naturally occur in day-to-day life among older adults.

(1) To describe how to design and implement a mobile-based EMA protocol for collecting psychosocial data, and to evaluate the feasibility and acceptability of this protocol among a sample of older Chinese.
(2) To test the overall between-person associations between the occurrence of and the satisfaction with social interactions with the average level of momentary affective states.
(3) To test the within-person temporal associations between the occurrence of and the satisfaction with social interactions and momentary affective states.
(4) To test the within-person lagged associations between momentary affective states and the subsequent level of satisfaction with social interactions.

1.3 Research significance

(1) Theoretical contribution

This study enhanced our knowledge about the dynamics interplay of social interaction and affective well-being regarding the following three aspects. First, this study highlights the importance of an examination of momentary features (both quantitative and qualitative) of social interactions that may change across situations throughout the day and the implications of these features for boosting or deflating positive and negative affective states. Second, our study was able to properly separate the between- and within-person effects of social interactions on affective experiences, which helps to clarify the relative importance of the long-term structural properties and the short-term process properties of social relations for shaping affective experiences. Third, this study tracked moment-to-moment social interactions and examined their concurrent and lagged relations with intra-individual variations in affective experiences. By doing so, our study was able to provide a nuanced picture of the temporal pattering in social relationships and affective experiences.

(2) Methodological contribution

This study developed a mobile-based EMA protocol for the use of collecting data of moment-to-moment experiences of social interactions and affective states. It provided one of the first description of the key issues that researchers need to know before designing and implementing a sophisticated EMA protocol with older people. It also provided the first evaluation of the full compliance patterns and subjective ratings of the feasibility and usability of this protocol in a sample of independently living urban seniors. By focusing on the methodological unknowns noted previously regarding the reliable use of smartphone-based EMA, our study may inform useful strategies during the design and implementation of mobile-based EMA to enhance older participants' adherence and satisfaction with the protocol.

Regarding the data obtained by adopting EMA methods, it has great potential for overcoming the recall bias inherent in retrospective reports and enhanced the ecological validity of the measures. Furthermore, the intensive longitudinal nature of the obtained EMA data allowed us to test a range of new research questions concerning the momentary within-person processes in later life, which was theorized to explain the affective benefits of social interaction but has been rarely explored to date.

(3) Practical contribution

Our research findings can inform the development of programs that aimed to promote healthier social environments for the maintenance and even betterment of affective well-being of older people. Also, practitioners (e.g., community mental health professionals, social workers, community consultants) may gain valuable insights from our study into the future development of momentary ecological interventions that incorporate mobile technology (such as self-monitoring technology and interactive technology) into psychosocial treatments. For example, informational support (e.g., remind subjects to engage in a specific behavior or reappraisal when confronted with a stressor) could be provided during the occurrence of situations that may trigger enduring negative affect (e.g., subjects encountered difficulty in recovering from exposure to a negative event) or before the foreseeable critical events. Also, combing these mobile techniques into psychosocial treatments provide opportunities for continuous self-monitoring and the provision of personalized interactive interventions, for example, once the individual become aware of his or her changes in emotional or behavioral patterns (e.g., increasing discrepancy between actual and desired affective states), she or he could initiate an interactive feedback or search for other types of support.

In the next chapter, an overview of the current literature on the study topic will be presented, followed by an articulation of the theoretical frameworks and hypotheses. In Chapter 4, we gave a detailed discussion of the methodology we adopted for the present study. In Chapter 5, the major results of this study will be presented. In Chapter 6, we will give a discussion on the results as well as the contributions and limitations of the current study.

Chapter 2 Literature review

The first section of this chapter discusses the extant research on the between-person relations of social interactions and affective experiences, with particular focus on how the quantitative (objective) and qualitative (subjective) aspects of social interactions contributed to affective well-being. In the second section, the existing research addressing the within-person covariation of social interactions and affective experiences was reviewed. In the third session, we gave an overview of the application of ecological momentary assessment (EMA) for the research on psychosocial experiences, and in particular the feasibility of this method to be used with aging populations. In the last section of this chapter, we gave a summary of the reviewed studies and also identified limitations remained in this literature.

2.1 Between-person Relations of Social Interactions and Affect

There has accumulated a considerable number of research of the between-person relations of social interactions and affective experiences (S. T. Charles & Carstensen, 2010; Pinquart, 2001). At the most general sense, individuals who have significant others and more frequent social engagement or interpersonal contact reported better affective outcomes (Berkman, 1995; Deindl et al., 2016). A growing number of studies also demonstrated the significance of the qualitative aspects of social relationships for shaping individual affective outcomes (Amieva et al., 2010; Fiorillo & Sabatini, 2011). Once the quality of interactions is considered (e.g., relationship satisfaction, perceived support), the quantitative aspects of interactions becomes less influential in explaining health and well-being outcomes. Thus, both the quantitative and qualitative features of social interactions warranted research attention for a fuller understanding of affective implications of social environment.

2.1.1 Quantitative features of social interactions and affective experiences

Extant research devoted to examining how the quantitative characteristics of individuals' social relations are related to affective outcomes. In this research, the quantity of social interactions was most commonly operationalized as the size of social network (e.g., the number of close ties such as spouse, family, friends), the frequency of social contact with different social partners (e.g., family, friends, neighbors, co-workers, or other acquaintances). The prevailing view in the research is that, having interactions with others (versus being socially isolated) makes people feel integrated as active parts of the society, thus contributes to individual well-being outcomes. This was supported by a substantial number of empirical studies suggesting that individuals who reported a larger social network size and more frequent social contact also tended to have more positive and less negative affective experiences (Fiori, Antonucci, & Cortina, 2006; Jose & Shanuga, 2015).

This research described above (especially those having prospective designs) have been valuable for demonstrating the overall significance of social interactions for affective well-being, however, it remains an inconclusive question regarding the remarkable variations in the strength, scope, and even the direction of the established relationships social interactions and subjective well-being outcomes in old age (Fiorillo & Sabatini, 2011). More frequent interaction with social partners is not always beneficial to older people and being embedded in a larger or more diverse social network (that has long been seen as a better indicator of social integration) may not always provide opportunities for getting needed support and accompany without cost. Especially for the older population, a larger social network may indicate the growing need for getting assistance from others because of declined independence, and the investment necessary for maintaining such a large network can also create pressure and burden, which may subsequently lead to worse emotional outcomes such as experiencing negative affect such as anxious and guilty more frequently (Schneider et al., 2013; Windsor et al., 2014).

Regarding the inconsistencies reported by studies examining affective implications of single aspects of social interactions (size of network, contact frequency), more recent researches started to use the individual-centered approaches to identify different types or patterns of social networks and examine their predictive utility of health and well-being outcomes. Across the studies focusing on the social network types of older people, researchers have established four robust typologies, including friend-intensive, family-intensive, diverse, and restrictive patterns (Fiori, Smith, & Antonucci, 2007). Importantly, these different patterns have been shown to exhibit significant impacts on emotional health and other indicators of subjective well-being (Litwin & Shiovitz-Ezra, 2011b; Park, Smith, & Dunkle, 2014). As stated by observers, these patterns are distinct from each other in terms of the proportion of different relationship types in certain type of network, suggesting that theses relationship types can function differentially in contributing to subjective well-being outcomes (Antonucci, Lansford, & Akiyama, 2001; Huxhold, Miche, & Schüz, 2014; Wilson, Harris, & Vazire, 2015).

Specifically, interactions with close family have been long regarded as the most important source older people may rely on for emotional support, however, these relationships may also create conflict, ambivalence and stress that subsequently result in unfavorable consequences. By gaining comfort and assurance, people can benefit from close social relationships (Fiori et al., 2006), but in parallel, it might be inevitably that these relationships often bring them pressures and conflicts (Cohen, 2004). Especially for the later adulthood, the feeling of being cared for and being loved can provide older individuals a sense of security and a feeling of belonging, whereas the need to rely on others for support (either financial, instrumental or emotional) may put pressure or even strain on supportive relationships. For example, the relationship between parents and adult children or the relationship between spouses has been reported as involving tension and ambivalence, which has been prospectively associated with adverse health and well-being outcomes for older persons (Holt-Lunstad et al., 2007; Losada et al., 2018; Ross, Guardino, Hobel, & Schetter, 2018; Walker, Luszcz, Gerstorf, & Hoppmann, 2011; Willson et al., 2003). It is not only support but also strain in these relationships can be significantly associated with relationship adjustment, affective well-being and risks for declined mental health outcomes (Bengtson, Giarrusso, Mabry, & Silverstein, 2002; Berry & Worthington Jr, 2001; Kira S Birditt, Miller, et al., 2009; Heid, Kim, Zarit, Birditt, & Fingerman, 2018; Lüscher & Pillemer, 1998; Yang et al., 2016).

Unlike family roles and relations that are socially defined and related to obligations, friendships are usually age companions, chosen by choice, and are typically operated under the principles and norms of equity and reciprocity. The current literature tended to assert that embedding in a friend-intensive network serves as an important source of companionship, mutual respect and emotional support, which could boost self-esteem and greater subjective well-being and protects against unhealthy emotional patterns such as depression and anxiety (Choi, Burr, Mutchler,

& Caro, 2007; Nguyen, Chatters, Taylor, & Mouzon, 2016). Especially for individuals at the latest life stage, friendship relations have been proposed to have a "buffering effect", which can act as alternative sources of support following the loss of role identity (e.g., spouse, employee) or the intimate family relationship in later years (for example, releasing the pain of bereavement) (Carr, Cornman, & Freedman, 2016; Rawlins, 2017; Young & Glasgow, 1998). Furthermore, researchers have pointed out that interacting with friends and family can have additive effects on affective well-being. For example, the family-focused social networks have been associated with lower mental well-being as compared to networks that include both family and friend relationships. Also, older people who perceived greater support from their family members alone tended to report lower levels of subjective well-being than those who perceived support from both their families and friends (Bengtson, Giarrusso, Mabry, & Silverstein, 2002; Fiori et al., 2006; Proulx et al., 2007).

2.1.2 Qualitative features of social interactions and affective experiences

Although the pattern-centered approach has significantly contributed to the literature by incorporating multiple aspects of social interactions, the differentiation of network patterns still relies on the quantitative (structural) features (e.g., network size, the total amount or frequency of social contact) of social relationships. More recently, an increasing attention has been given to the quality of social interactions, which has been suggested being more powerful in predicting affective well-being outcomes than the quantity of social interactions in specific relationship domains (Antonucci et al., 2001). Once these qualitative aspects of interactions were considered, the quantitative aspects of interactions become less influential in explaining affective outcomes. For instance, Fiorillo and Sabatini (2011) found that when relationship satisfaction was considered, which exhibited a strong positive impact on subjective well-being, the magnitude of the positive relationship between engagement in frequent social interactions and subjective well-being was declined. Similar research findings have also been revealed elsewhere (Cooney & Dykstra, 2013; Henriksen, Torsheim, & Thuen, 2015).

Following the central assumption that relative to loneliness (also referred to subjective isolation or perceived isolation), being connected or integrated may result in better well-being status, most of the empirical researches tended to capture the quality of social interaction by a unidimensional continuum that ranged from the perception of disconnection to connectivity. Researchers have used different terminology to indicate the perceived quality of social interactions (perceived social support from important others, perceived as being cared for and valued by others) or sometimes referred to as social cohesion (for example, perceived others as friendly and trustworthy). Yet, this widely used approach has been criticized for addressing the qualitative aspect (or subjective experiences) emergent from social interactions in a recessive and indirect manner. In particular, social interactions that were characterized by negative features (e.g., interpersonal arguments and confrontation; having poor communication quality) tended to receive much more research attention, and have been prospectively associated with emotional distress and other high-arousal negative affective states such as anxious and angry (Martinez-Corts, Demerouti, Bakker, & Boz, 2015; Turner, Cobb, Gratz, & Chapman, 2016; Wickham, Williamson, Beard, Kobayashi, & Hirst, 2016).

A growing number of studies recognized that rather than being treated as a unidimensional scale, the operational definition of the quality of social interaction should embrace both positive and negative aspects, given the fact that these aspects may be appraised by a person at the same time (Fingerman et al., 2004). Generally speaking, it has been widely believed that positive aspects of social contact (e.g., warm and aggregable communication) are related to enhanced well-being of older people, and in contrast, negative aspects of social contact (e.g., disagreement, conflict, uninvolved, un-reciprocal) could lead to adverse subjective well-being outcomes (H. J. Lee & Szinovacz, 2016; von Humboldt, Leal, & Pimenta, 2015; R. S. Wilson et al., 2015; Windsor et al., 2014).

More specifically, there have been two competing hypotheses in the literature. One is called 'domain-specific effect', which asserts that the positivity and negativity of social contact are equally important for affective well-being and exert a valence-specific effect on affective well-being, with the positivity of social contact mainly influencing positive affect and the negativity of social contact mainly influencing negative affect. Evidences supporting this assumption included that when the level of interaction positivity remains the same, the level of interaction negativity predicted more negative affect but did not influence positive affect. on the other side, if the level of interaction negativity was held constant, the level of interaction positivity only mattered for the experiences of positive affect but not for negative affect (Hawkley, Preacher, & Cacioppo, 2007; Hofmann, Finkel, & Fitzsimons, 2015; Rook, 2001); Steptoe (2000).

On the other side, a stream of research tended to support the 'crossover effect' hypothesis, which suggested that the effects of negative social exchange may be sufficient to spread across the emotional realm, serving to inhibit positive emotional states and also trigger negative states in a persistent manner. This view was also supported by several studies with longitudinal designs. For example, one study reported that when the positive exchange remained the same, the amount of negative exchanges were prospectively associated with both positive and negative mood at follow-up (Newsom, Nishishiba, Morgan, & Rook, 2003). However, this enduring "cross-domain" effect has not been observed for the baseline amount of positive social exchanges as it was unrelated with either positive or negative affect at follow-up. This inconsistency may reflect the different temporal patterns of positive and negative social interactions in relation to affect, as the effects of interaction negativity tended to be more persistent than that of the interaction positivity (Kira S Birditt et al., 2018; Hertel, 1999; Newsom et al., 2003; Okun & Keith, 1998).

On the methodological side, the above studies on the impact of social exchange rely on the retrospective measure that requires the subjects to recall their experiences over a certain time frame (recall the number of interpersonal conflicts experienced over the past month; recall the degree to which you felt uncomfortable with the interaction with your spouse). Unlike these retrospective designs, other studies adopted diary methods (or day reconstruction method) that asked their participants to report their experiences throughout a given day. Their results suggested that individuals who reported more unpleasant events or interpersonal stressors also experienced worse subjective well-being and more frequent negative affect. Consistent with the "crossover effect" hypothesis, the impact of negative social exchanges on affective responses has been shown as more pronounced than that of positive interaction. Yet, it is also possible that the effect of negative features of social exchange has been over-estimated because most of these dairy studies tended to focus on negative affective states as study outcomes (Bolger, DeLongis, Kessler, & Schilling, 1989; Ingersoll-Dayton, Morgan, & Antonucci, 1997; Peeters, Buunk, & Schaufeli, 1995; Rook, 1984; Rook, Thuras, & Lewis, 1990; Vittengl & Holt, 1998b; Zautra, Burleson, Matt, Roth, & Burrows, 1994).

2.2 Within-person Covariation of Social Interactions and Momentary Affect

During the past three decades, there has been an accumulation of research addressing the within-person covariation between everyday social interactions and momentary affective experiences during adulthood. These studies not only focused on the correlations between specific features of social interactions and the intra-individual differences in affective states, but also started to test the temporal sequencing of momentary social and affective experiences. Moreover, they have examined a variety of between-person level factors (e.g., psychological traits) that moderated the within-person covariation of social interactions and affect.

2.2.1 The occurrence of social interactions and momentary affect

2.2.1.1 Operationalization of occurrence of social interaction and momentary affect

Early studies devoted to understanding within-person covariations of social interactions and momentary affect primarily focused on the occurrence of social situations. These studies showed some variations in terms of their operational definitions of occurrence of social interaction. In this literature, there is a consensus that two criterions are useful for clearly determining an operational definition of social interaction. The first criterion is that this a situation involving at least two persons (e.g., one subject and one partner). Many researchers also pointed out that this situation could either be an interaction where persons were physically involved, or an interaction where persons interacted using phone calls, video, and social media). The second criterion is that this is a situation where the behaviors of each person were mutually influenced regardless of whether the influence is direct or not. However, in our reviewed studies of the within-person relations between social interactions and affect, only a few numbers have incorporated both criterions. Specifically, seven studies only considered the first criterion by asking participants to report on the presence of other persons during a certain time period. In thirteen studies, the occurrence of social interaction was defined by a face-to-face interaction with other people that lasted for a certain number of minutes or longer (e.g., 5 to 10 minutes). By doing so, the operational definitions incorporated both two criterions mentioned above but excluded the situations where people interacted electronically. Only one study (Alshamsi et al., 2016) fully incorporated the two criterions by including both face-to-face interactions and digital (e.g., video chat) interactions. See Table 2.1 for an overview of the operationalizations of the occurrence of social interaction and momentary affective states.

These reviewed studies also differed significantly from each other in their operationalizations of momentary affect measures. In general, most studies kept with the central assumption underlying a two-factor model of the basic of affect, that is, positive affect (PA) and negative affect (NA) should be treated as two independent dimensions and may have different correlates (Posner, Russell, &

Peterson, 2005). Therefore, there is a consensus in our reviewed studies that PA and NA may be differentially influenced by the characteristics of social interactions. Specifically, the majority obtained both measures of the concurrent levels of PA and NA over time across situations. Typically, these studies were based on uni-polar scales with a number of selected items (for the original work, see Watson, Clark, & Tellegen, 1988). Three studies (Hepp et al., 2017; Vella et al., 2012; Kashdan et al., 2014) focused on either the concurrent level of PA or NA.

While the studies mentioned above focused more on the valence dimension of affect, there were eight studies explicitly taking into account both the valence and arousal dimensions to distinguish different types of affective states. Three studies distinguished high aroused PA from low aroused PA, and high aroused NA from low aroused NA. In five studies, it had been explicitly stated that the operationalization of affect in their studies was based on the idea that valence dimension (e.g., a continuum ranging from very unpleasant to very pleasant*)* and arousal dimension (e.g., a continuum ranging from very sleep to very energetic) can distinguish specific types of affective states. Table 2.1 gave a summary of the operationalization used by different studies in details.

This literature review was conducted by searching the online databases PsycInfo, PubMed, and Medline for relevant publications up to July 2018, using the following string of search terms: (momentary assessment OR diary OR experience sampling OR ecologic momentary assessment OR ecological momentary assessment OR event-contingent recording OR micro longitudinal OR momentary measures) AND (affect OR affective OR emotion OR mood) AND (social OR interpersonal). Filters have been used to limit our search results to articles published in peer reviewed journals in the English language. We used the following criteria to select the studies: (1) the study reported original empirical findings based on quantitative analysis of EMA data, (2) the study sample was restricted to adults aged 18 or above; (3) the study included state-level measurements of social interactions and momentary affect obtaining from the EMA method. In total, 28 independent studies (from 30 articles) were identified by our reviewers as eligible according to the study eligibility criteria. Among these 28 studies, 23 studies provided operational definition of occurrence of social interaction as well as momentary affect 9as shown in Table 2.1).

Table 2.1 The operationalization of occurrence of social interaction and momentary affect

Authors	**Occurrence of social interaction**	**Momentary Affect**
Brandstätter (1983)	Presence of any social partners; Types of social partners;	Mood scores (negative, indifferent, positive)
Burgin et al. (2012)	Presence of social partners;	Positive affect measures Negative affect measures
H. Chui, Hoppmann, Gerstorf, Walker, and Luszcz (2014)	Presence of any social partners; Types of social partners	Positive affect measures Negative affect measures

Larson et al. (1985; 1986)	Presence of any social partners; Types of social partners	Valence; Arousal
Pauly, Lay, Nater, Scott, and Hoppmann (2017)	Presence of other persons (including situations where the other persons were present electronically)	High aroused positive affect Low aroused positive affect High aroused negative affect Low aroused negative affect
Torquati and Raffaelli (2004)	Presence of other persons;	Valence; Energy; Connection
Wichers et al. (2015)	Presence of other persons;	Positive affect measures Negative affect measures
Lucas, Le, and Dyrenforth (2008)	Presence of others and communication	Positive affect measures
Alshamsi, Pianesi, Lepri, Pentland, and Rahwan (2016)	Face-to-face or digital interaction where the subject talked with someone.	High aroused positive affect Low aroused positive affect High aroused negative affect Low aroused negative affect
Bernstein et al., (2017)	Face-to-face interaction that lasted for at least 5 minutes.	Positive affect scale Negative affect scale

Côté and Moskowitz (1998)	Face-to-face interaction with someone;	Positive affect measures Negative affect measures
Flory et al., (2000)	Face-to-face interaction with someone prior to the beep;	Positive affect measures Negative affect measures
Granholmn et al., (2013)	Face-to-face interaction with someone prior to the beep;	Three items from PANAS (happy, sad, anxious)
Hawkley et al. (2007)	Face-to-face interaction with someone;	20-item PANAS
Hepp et al. (2017)	Face-to-face interaction with someone;	Negative affect measures
Sadikaj et al., (2010, 2011, 2013)	Face-to-face interactions lasting over 5 minutes;	Positive affect measures Negative affect measures
Timmermans, Van Mechelen, and Kuppens (2010)	Face-to-face interactions lasting over 5 minutes;	Valence; Arousal
Vella et al., (2012)	Face-to-face interactions lasting over 5 minutes;	Hostile mood

Vogel et al., (2017)	Face-to-face interactions lasting over 5 minutes;	Valence; Arousal
Wang et al. (2014)	Face-to-face interactions lasting over 5 minutes;	Valence; Arousal
Gillian M Sandstrom and Dunn (2014)	Greeted someone in person, regardless of the interaction length)	20-item PANAS
Jeffrey and Holt (1998)	Not defined	20-item PANAS
Sandstrom et al. (2017)	Being in a social situation (family/friend's house, or restaurant/cafe/hub)	High arousal positive affect Low arousal positive affect High arousal negative affect Low arousal negative affect

2.2.1.2 Concurrent relations: the occurrence of social interactions and momentary affect

Early studies concerning the within-person covarion of social interactions and momentary affect focused on the presence of other people, in other words, examining whether people may experience different affective states when they were with social partners, relative to when they were alone (e.g., solitude). At a rough sense, their results indicated that being with other people rather than being alone benefitted individuals by coupling with higher level of positive affect and lower level of negative affect (Brandstätter, 1983; Watson, Clark, McIntyre, & Hamaker, 1992). For example, in one of the earliest daily diary studies, Larson et al., (1985) reported that relative to being alone, participants tended to be happier when they were with others. While the study above was conducted with a sample of community older people, the authors extended the results in their later study using a sample of residents living in nursing facilities, suggesting that older residents' positive affect was positively correlated with their contact with their family members.

In a more recent study (Pauly et al., 2017), the within-person effects of being with others has been interpreted in the context of valence-specific and arousal-specific affective implications. Using a sample of community adults with a wide range of age (between 21 and 81 years old), researchers found that being alone was associated with unfavorable affective experiences, in terms of decreased high arousal PA such as happy or excited and increased low arousal NA such as sad and sleepy. However, time spent alone was also associated with increased low arousal positive affect, pointed out the fact that being alone also offer some affective benefits during old age because it can enhance low arousal positive states such as calmness and relaxation (Kira S. Birditt, Manalel, Sommers, Luong, & Fingerman, 2018; Jennifer Christina Lay, 2018; Jennifer C Lay, Pauly, Graf, Biesanz, & Hoppmann, 2018; Long & Averill, 2003; Matias, Nicolson, & Freire, 2011).

A number of studies paid attention to examine whether interacting with different social partners have differential influences on affective well-being. First, the beneficial effects of being with friends have been demonstrated by diary studies and experience sampling studies (a term interchangeably used with "ecological momentary assessment" in the literature). For example, time spent with one's friends have been consistently linked with more positive affective experiences while the impact on negative affective experiences is minimal. For example, in a one-week experience sampling method study, older people reported greater happiness (above the individual's own means) when they were with friends, as compared to when they were with other types of social partners including spouse and family members. This result highlighted again the affective benefits of being with friends for older people (R. Larson et al., 1985, 1986).

Second, being with one's spouse (romantic partner) has been suggested as being not necessarily benefit affect and sometimes may even be coupled with unfavorable affective experiences, especially for a subgroup of individuals with certain personality traits or symptoms. For example, Chui et al. (2014) examined the within-person coupling of momentary affect with the presence of social partners with different relationship types (e.g., spouse, other family member, friend, formal care giver, service provider and others) in a sample of the oldest-old. Unlike the presence of friends and family that has been consistently associated with more positive affective experiences, their results suggested that the presence of one's spouse failed to have an impact either on positive or negative affect among the oldest-old. Moreover, when examined within person-specific characteristics (gender, health), the effect of being with spouse became significant in predicting negative affective experiences. In specific, older women and older people with more chronic illnesses reported more NA and fewer PA when they were with their spouses, relative to when they were with other types of partners.

In addition to the different relationship types, another straightforward way to quantify the uniqueness of people who we interact with is to distinguish between the close and peripheral social partners. Interacting with close ties (familiar) has been correlated with more positive affect in several studies. For example, Gillian M Sandstrom and Dunn (2014) examined whether affective states is related to the number of daily interactions people had with partners from close ties and with partners from peripheral ties. They examined the same research question in two distinct samples, one focused on university students and another for community adult members. In both samples, more interactions with strong ties than typical (the person's average level) have been found to be related to greater affective well-being. The positive correlation between interactions with familiar people and positive valence has also been documented in a study of university students (Jeffrey et. al, 1998), and a study of community adults (Vogel et al., 2017). In the latter one, Vogel et al. (2017) also examined the relationship between partner familiarity and arousal, where unexpected nonlinear associations were found that arousal was highest during interactions with medium familiar partners. In addition, the contribution of peripheral ties for individual's day-to-day affective well-being was reported by Sandstorm and Dunn (2014). In the sample of university students, more weak ties interactions than typical were related to greater affective-well-being, and such beneficial effect of additional interactions with weak ties was more pronounced during days when people have fewer interactions with strong ties (relative to their average level). This result highlighted the importance of peripheral ties in contributing to individual's affective well-being, especially in situations where strong ties interactions were absent.

2.2.2 The subjective appraisals of social interactions and momentary affect

2.2.2.1 Operationalization of the subjective appraisal of social interactions

In this review, we reviewed a total number of 17 studies addressing the within-person covariation of the subjective appraisal of a certain social interaction and momentary affect. The selection of the 17 studies was based on the literature review described in section 2.2.1, which provided a total of 28 studies addressing the link between state-level social interactions and affect among adult population (aged 18 and over) using EMA methods. These studies showed great variations in terms of how the subjective appraisals of social interactions were defined and assessed. Table 2.2 presented an overview of the measures of the appraisal of social interactions used by reviewed studies. Six studies used single items with a bipolar response scale to measure the subjective perception of the social interaction, for instance, the self-rated pleasantness of or satisfaction with an ongoing interaction (Bernstein et al., 2017; Hofmann et al., 2015; Flory et al., 2000; Jeffery and Craig, 1998b; Wichers et al., 2015), or the self-rated importance of an interaction (Vogel et al., 2017). Three studies (Hepp et al., 2017; Hahn et al., 2013; Jeffery and Craig, 1998a) implicitly defined a social interaction as either positive or negative by asking individuals to report on a list of pre-defined types of social interactions, for example, fun or active interaction was treated as positive interactions while arguing or confronting was considered as negative interactions. In contrast, four studies assessed multiple parameters of social interactions reflecting the fact that individuals may experience a blending of positive and negative feelings when communicating with specific social partners (Hawkley et al., 2007; Kashdan et al., 2014; Vella et al., 2012).

The other four studies differed significantly from the studies above. Instead of asking about the features of the ongoing social interactions, these studies assessed participant's subjective perceptions of their own behaviors or the partner's behaviors (Timmermans et al., 2010; Sadikaj et al., 2007, 2010, 2011, 2013; Wang et al., 2014). They are consistent in their measurement of interpersonal perceptions, which were categorized into two basic dimensions of psychological inputs. The dimension of agency referred to whether the behaviors of a subject or a partner was perceived as dominant and assured versus submissive and unassured; and the dimension of communion considered whether the behaviors of a subject or a partner was perceived as friendly and aggregable versus cold and quarrelsome.

Table 2.2 The Operationalization of the Subjective Appraisals of Social Interactions

Author	Measures of the appraisal of social interaction
Jeffery and Craig (1998a)	Features of SIs including fun or active; arguing/confronting; necessary or informational

Hepp et al. (2017)	Interpersonal stressors (i.e., rejection, disagreement)
Hahn et al. (2013)	Social conflict and interpersonal stressors
Jeffery and Craig (1998b)	Quality of communication (i.e., difficult-smooth; unpleasant-pleasant)
Bernstein et al. (2017)	Pleasantness of SIs;
Wichers et al. (2015)	Pleasantness of the company;
Hofmann et al. (2015)	Momentary relationship satisfaction;
Vogel et al. (2017)	Importance of the SIs
Flory et al. (2000)	Subjective rating of the interaction as positive, neutral, negative, or an argument;
Granholm et al. (2013)	Positivity of the SIs (successful, worthwhile, likeable)
Kashdan et al. (2014)	Positivity of SI (warm/aggregable, intimacy, enjoyment, influence)
Hawkley et al. (2007)	Positivity (i.e., comfortable, intimate, involved) and Negativity of SIs (i.e., cautious, conflicted)

Vella et al. (2012)	Positivity (agreeableness and emotional express) and Negativity of SI (being treated badly, being interfered)
Côté and Moskowitz (1998)	Subject's Agency (assured versus unassured; dominant versus submissive) and Communion (warm versus cold; agreeable versus cold-quarrelsome)
Timmermans et al. (2010)	Subject's Agency (assured versus unassured; dominant versus submissive) and Communion (warm versus cold; agreeable versus cold-quarrelsome)
Wang et al. (2014)	Partner's Agency (assured versus unassured; dominant versus submissive) and Communion (warm versus cold; agreeable versus cold-quarrelsome)
Sadikaj et al. (2007, 2010, 2011, 2013)	Partner's Agency (assured versus unassured; dominant versus submissive) and Communion (warm versus cold; agreeable versus cold-quarrelsome)

2.2.2.2 Concurrent covariation between appraisals of social interactions and momentary affect

Early studies concerning the quality (subjective evaluation) of social interactions tended to characterize social interactions as either primarily positive or negative, and examined the relations between affective experiences with positive and negative interactions separately. For example, one of the earliest studies using experience sampling method focused on five types of social interaction among college students: necessary or informative, interesting or positive, providing help and support, receiving help and support, argument or confrontation. The first three types were considered as positive interaction while the last two types were regarded as negative (Jeffery and Craig, 1998a). Their result tended to support the "domain-specific" hypothesis discussed in former sections (e.g., for more details about this hypothesis, see section 2.1.2), that is, qualitatively distinct types of social interactions exerted a valence-specific effect on affect, with positive interactions being concurrently coupled with positive affect while negative interactions being concurrently coupled with negative affect. Similarly, some studies focusing on affective response to interpersonal stressors have revealed that within individuals, these interpersonal stressors (considered as negative social interactions) strongly predicted aroused negative affect but showed minimal impacts on positive affect. Similar findings supporting this view that negative interactions triggered negative affect have been reported in samples of married and unmarried older people (Hahn et al., 2013), community adults (Machell, Kashdan, Short, & Nezlek, 2015; Nezlek, 2005), as well as in samples of clinical populations such as those patients with specific psychiatric disorder (Hepp et al., 2017).

Unlike the studies above that assumed a certain interaction as either positive or negative, some studies used single items with a bipolar response scale to measure the subjective perception of the social interaction. Generally, these studies tended to suggest that the level of satisfaction or pleasantness of an ongoing social interaction was predictive for the concurrent levels of momentary affect (e.g., Bernstein et al., 2017; Hofmann et al., 2015; Vogel et al., 2017; Wichers et al., 2015). Taking a recent experience sampling study as example, in which the participants were asked to report their state-level relationship satisfaction with tie partner for multiple times per day during a one-week period (Hofmann et al., 2015) Their results suggested that when participants reported a higher level of relationship satisfaction (above the typical level of themselves), they experienced higher level of positive affect and achieved better progress of their daily goal purist (e.g., emotional regulation). Another example is the study conducted by Flory et al., (2000). In a sample of 100 middle-aged persons, the participants provided concurrent reports of their subjective rating on a certain interaction that happened within ten minutes as positive, neutral, negative, or an argument. The results suggested that greater exposure to negative social interactions than usual were related to decreased PA and increased NA in the entire sample, while people with particular characteristics were found to be more vulnerable to such effects. Similarly, Bernstein et al. (2017) examined concurrent associations between employee's affective states and their self-rated pleasantness of social interactions, with a seven-point scale (0 = unpleasant and 6 = pleasant). They found that individuals experienced greater happiness and fewer sadness during interactions they rated as more pleasant than typical (the participant's own average-level).

While the reviewed studies above tended to focus on either positive or negative aspects of social contact, it has been increasingly recognized that in real life, people often engage in interactions characterized by a mixture of positive and negative parameters, for example, an interaction could be aggregable and cold at the same time (Holt-Lunstad et al., 2007). Therefore, some studies have begun to obtain direct measurements of subjective appraisal (or perception) of social interactions to capture both the positive and negative aspects of social interaction at the same time. Two studies examined the separate influence of interactions perceived as negative and interactions perceived as positive on affect. Vella et al. (2012) obtained measurement of multiple characteristics an ongoing concurrent social interaction for every 45-minute, where negative interaction variables were based upon participants' responses to three of these characteristics including "someone treated you badly", "someone interfered with your efforts", and "someone was in conflict with you"; and similarly, positive interaction variables were based upon items such as "whether the interaction is agreeable" and "someone expressed care/concern for you". They found that the higher level of negativity of social interactions predicted elevated hostile mood, whereas the level of positivity of social interactions did not relate to daily hostile mood. This result that only negative social interactions mattered for negative affect tended to support the hypothesis about the domain-specific effects of social interactions on affective well-being.

Somewhat contradictory to the former study, the study conducted by Hawkley et al. (2007) indicated that both positive and negative features of social interactions were predictive for the concurrent experiences of positive and negative affect. In this study, the immediate assessment of the quality of social interactions were based upon participants' responses to 16 adjectives (scaled from 1 = not at all to 5 = very), of which responses to 8 positive aspects of interaction were averaged to generate an index of interaction positivity (i.e., comfortable, supported, involved, etc.) and the other responses to 8 negative aspects were averaged to create an index of interaction negativity (i.e., conflicted, dishonest, etc.). Their results suggested that higher score on the interaction positivity scale significantly predicted the experiences of more PA and fewer NA, and by contrast, higher score on the interaction negativity scale significantly predicted the experiences of more NA and less PA. This result tended to support the assertion that both positive and negative features of interactions are important predictors of momentary affective outcomes.

Other four studies tended to focus on subjective perceptions of the subject's own or the partner's behaviors. Two studies examined individual's perceptions of their own behaviors during interactions. More communal behaviors than usual were found to be associated with positive valence. Specifically, in the 20-day event-contingent sampling study of community dwelling adults (Côté & Moskowitz, 1998), participants reported higher valence when they perceived their behaviors as more communal (more agreeable and less quarrelsome). This within-person coupling of higher communion and higher valence was replicated in a 14-day signal-contingent study of university students (Timmermans et al., 2010), where higher pleasantness was found to be associated with social interactions perceived as higher in communion (above the person's average level). On the other side, inconsistent results emerged on the association between agentic behavior and affect, of which Côté & Moskowitz (1998) reported that participants experienced higher valence when they considered that they behaved at a more agentic manner than they typically did, whereas Timmermans et al., (2010) suggested that there was only a weak association between valence and agentic behaviors. Furthermore, Timmermans et al., (2010) also examined the within-person coupling of arousal and agentic behavior, and higher arousal was found to be strongly associated with subject's more agentic behaviors at both the between-and within-person level.

The behaviors of the interacting partner (perceived by the subject) also played a role in influencing affective experiences. When the partner was perceived with a higher level of communion, the subjects typically reported more favorable affective responses in terms of increased valence (Wang et al., 2014). For example, Sadikaj et al., (2011) conducted a study focusing on individuals' affective reactivity to partner's communal behavior using a community sample with a wide range of age (between 20 and 70). They found that reduced communal behavior may act as triggers for increased NA, while increased communal behavior may initiate subsequent PA. These effects have been found to be present in a various type of relationships (such as romantic partner, friends, co-workers, acquittance), with the most pronounced effect was found when the person was interacting with a romantic partner. In addition, the study reported by Wang et al. (2014) indicated that higher level of partner's agency was a strong predictor of heightened arousal. It has been explained that under circumstance where the partner behaved in a very dominant pattern, the subject may need to make more efforts to lead interpersonal competition to gain dominance, which would be likely to result in a heighted level of the subject's arousal.

2.2.3 Time-lagged relations of social interactions and momentary affect

2.2.3.1 Appraisal of social interactions and subsequent level of affect

We identified three studies addressing the lagged relations between the appraisal of social interactions and the subsequent level of momentary affect. The first study reported that the pleasantness during prior interaction served as a significant predictor for positive affect during subsequent interaction, when accounting for positive affect and other quantitative features of prior interaction (Kashdan, Yarbro, McKnight, & Nezlek, 2014). However, this study was limited by the mere focus on the level of positive affect, thus did not provide information about whether the pleasantness of a prior social interaction predicted the level of negative affect at subsequent time point.

Another study obtained self-reported data on the positive and negative features of an ongoing social interaction for every 45 minutes throughout a day (Vella et al., 2012). According to the time-lagged analysis, only negative interaction was predictive for elevated hostile mood at the subsequent time point, while no significant time-lagged effect of positive interactions was found on subsequent hostile mood. This result tended to support the hypothesis about the "domain-specific effects" of social exchange on affective well-being. Also, it is possible that the influence of negative interactions may be more persistent than positive interactions. However, this study was limited by only focusing on hostile mood (one type of negative affect), the results can hardly reflect the differing nature of positive versus negative features of social interactions in terms of their influences on affective-well-being.

Unlike the former study, Hawkley et al. (2007) have linked both positive and negative features of social interactions to positive and negative affect. In this study, the time-legged models indicated that the positivity of interaction index significantly predicted more PA and less NA in the subsequent assessment, while the negativity of interaction index significantly predicted more NA and less PA. Again, their result lend support to the assertion that both positive and negative features of interactions are important predictors of momentary affective outcomes.

2.2.3.2 Affective states predicting subsequent features of social interactions

While most studies within this review largely interpreted their findings (correlations or time-lagged relations) to suggest that social interactions exhibited an impact on affect, we identified two studies investigating whether prior levels of momentary affect predicting subsequent social interactions. The first study (Wichers et al., 2015) examined the time-lagged relationship between current affect valence and the occurrence of similar social interactions in the near future using a relatively large sample of 610 female adults. The results suggested that positive affect associated with social contexts being perceived as pleasant increased this person's likelihood of engaging in similar social contexts in the near future, and on the other side, negative affect emergent from social contexts being perceived as unpleasant lead people to avoid subsequent engagement in similar social contexts. In another study mentioned before (Hawkley et al., 2007), the prior level of momentary affect was found to be predictive for qualitative features of subsequent social interactions. The results suggested that negative affect predicted less positivity and more negativity of subsequent interactions, and positive affect was predictive for higher positivity of subsequent interactions, but not for negativity of subsequent interactions. Given the strength of these time-lagged effects relating interaction quality and affect, the authors suggested that the reciprocal influence between negative affect and interaction negativity may be especially persistent (e.g., the impact continues to be significant even over two measurement occasions that separated for 2 hours).

2.2.4 Individual differences in the within-person covariation between social interactions and momentary affect

Many studies in this review examined the role of individual differences in explaining the variability in affective states in relation to daily social interactions. Age, social connectedness, mood disorder symptoms (e.g., depressive symptoms) and other trait-level psychological measures (e.g., personality traits, attachment styles) appeared to be significant predictors that have been reported by at least two studies. Overall, these findings bring attention to the necessity of considering individual differences while studying the dynamic intra-individual processes involving behavior and affective states.

2.2.4.1 Age

Age differences have been found in the within-person relation between social interactions and affect. Older people tended to be more immune than younger people to being alone and reported both higher levels of valence and arousal during interactions with family members than with non-family members. Using a lifespan sample, Pauly et al. (2017) found that with greater age, people tended to report less unfavorable affective states (decreased highly-activated PA and increased lowly-aroused NA), and more favorable affective states (increased low arousal PA) when they were alone. Given that this study made an explicit distinction between high and low arousal PA, as well as high and low arousal NA, these findings may provide preliminary evidence of the differential nature of valence-specific and arousal-specific effects of being alone in old age. Also using a lifespan sample, Vogel et al. (2017) reported that when compared with younger adults, older adults were less affectively aroused and reported less PA during interactions with non-family partners (e.g., co-workers). They also tended to be more affectively aroused during interactions perceived as important and meaningful. Taking the lens of several lifespan perspectives on emotional regulation and motivational goals, older people apprising momentary experiences of being alone less negatively might be explained by their enhanced emotion regulation skills and heightened focus on current positive experiences; they drawing more benefits from interactions with family members can be understood in the context of several life-span theoretical perspectives on emotional development (for example, socio-emotional selectivity theory), which predicts that older people selectively narrow their social life into those more predictable interactions with familiar partners to maintain emotional well-being (Laura L Carstensen et al., 2003).

2.2.4.2 Social connectedness

Social connectedness and loneliness may provide distinct advantages and disadvantages in individuals' social functioning and emotional well-being. Chui et al. (2014) found that momentary solitude was associated with more frequent negative affect, and this association was more pronounced for people with a higher score in trait-level loneliness than people with a lower score. For these lonely people, it was only the presence of his or her spouse was related to more frequent experiences of PA, suggesting the unpleasant affective experiences associated with being alone cannot be eased by the presence of social partners other than one's spouse. Further, people higher in loneliness have been found to report lower PA and higher NA on average, as well as less positivity and more negativity in interaction quality. Conversely, social connectedness predicted higher PA and lower NA, and more positive and less negative qualities in social interactions. But these effects were only significant in the between-person level as loneliness/social contentedness did not account for the strength of the relationship between fluctuations in affect and concurrent interaction.

2.2.4.3 Mood disorder symptoms

First, trait depression had a pervasive influence on daily affect, in that people who reported more frequent depressive symptoms also reported higher NA and lower PA than their heathier counterparts with fewer depressive symptoms throughout the study period (Flory et al., 2000). Moreover, a significant interaction between trait depression and negative social interactions for NA was found. This means that the relationship between negative interaction and increased NA was stronger for individuals who reported more depressive symptoms than for individuals who reported lower depressive symptoms. Second, Vella et al. (2012) demonstrated the role trait hostility played in shaping the magnitude of the relationship between momentary hostile mood (e.g., frequency) and social functioning. Hostile individuals (trait) not only reported more momentary hostile mood and negative social interactions, but also demonstrated greater affective reactivity to negative social interactions. Third, the role of trait social anxiety (SA) has been examined in a sample of university students (Vittengl & Holt, 1998b). Relative to the low SA group, the high SA group reported higher NA and lower quality of communication during social interaction and tended to experience more heightened NA when interacting with unfamiliar people.

2.2.4.4 Other trait-level psychological measures

There were some studies examining the role of personality traits for the within-person covariation between social contact and affective experiences. Neuroticism has been associated with more unfavorable affective experiences during daily social interactions. Agreeableness has also been related to patterns between interpersonal behaviors and affect, in that people higher in agreeableness reported more favorable affective responses than those less aggregable people when engaging in an agreeable interaction, but also reported less favorable affective responses when the interaction was perceived as lacking agreeableness (e.g., conflict, argument). In brief, these studies concluded that people higher in Agreeableness may be more emotionally sensitive to dynamics in social situations (H. Chui et al., 2014). In addition, studies consistently reported that extraverts engaged in social interactions more frequently and experienced more PA than did introverts. However, only one study (Lucas et al., 2008) evidenced that Extraversion moderated the relationship between social interactions and affect as extraverts experienced more pleasant affect than introverts when engaging in agreeable behaviors.

In addition to personality traits, some studies examined the role of attachment styles for explaining the within-person coupling of social interactions and affect. Relative to individuals with secure style, individuals with insecure style were more emotionally vulnerable to daily experiences of being alone bas they reported significantly lower levels of PA and energy and more extreme NA than secure individuals when they were alone (Torquati & Raffaelli, 2004). The within-person coupling of NA and partner’s lower communion (below the person’s average level) seems to be more pronounced for individuals higher on attachment anxiety (Sadikaj, Moskowitz, & Zuroff, 2011), which has been interpreted as signifying a failure in individuals' pursuit of acceptance and nurturance. In parallel, such within-person coupling was found to be weaker for individuals higher on attachment avoidance, suggesting less emotional distress experienced by attachment-avoidant people following others' negative behavior.

2.3 Ecological Momentary Assessment for Studying Real-life Experiences

The "Ecological Momentary Assessment" (EMA) is an important research method for examining temporal patterning in social interactions and affective states in real-life settings. In this section, we start with a brief overview of the historical development and contemporary use of EMA, followed by an introduction of the key methodological aspects of EMA. Then, we highlighted the benefits EMA studies provided for investigating the within-person process involving external social events and internal affective states. Third, we discussed the feasibility issues of EMA-based studies in aging population.

2.3.1 The historical development of EMA as a data collection method

The term "EMA" was first proposed in 1994 by Stone and Shiffman, which aimed to bring a range of diverse methods under a common methodological framework. This framework serves as a foundation where researchers and practitioners can select appropriate methods based on their targeted population and the purpose of their research or interventions. According to Shiffman, Stone, and Hufford (2008), it is encouraged to consider EMA as a collection of methods that share the following features: (1) data are obtained in natural settings rather than in the laboratory or clinic settings. By doing so, EMA provides data that are rich in contextual information and largely enhance the generalizability of the results to real life situations (higher ecological validity). (2) collect immediate reports of the subject's current experiences (events, behaviors, cognition, affect). By doing so, the EMA aimed to reduce recall bias that are inherent in many traditional retrospective assessments. (3) repeated measures were obtained over time and situations from the same person. This is the key feature that enables EMA to capture within-subject changes in behaviors with wide variations over time and across contexts, and to reveal and the mechanisms underlying behavior (Schwarz, 2007).

The early development of EMA draws on several methodological traditions (for reviews, see Beal, 2015; Brose & Ebner-Priemer, 2015; Shiffman et al., 2008) including daily diary method (sometimes day reconstruction method was also included as an alternative), self-monitoring techniques, ambulatory physiological monitoring, and the collection of medication compliance data. Paper-based daily diary is the earliest method used (still in use today), which was systematically developed since the 1940s for the use of clinical research. Self-monitoring of behaviors and clinically relevant events also has a long history in behavioral medicine research (Snyder, 1979). In the area of psychosocial behavior research, the Rochester Interaction Record is an important work contributing to the development of EMA (Reis & Wheeler, 1991; Wheeler, Reis, & Nezlek, 1983), which was designed to collect data about every social interaction occurred to the subject during their daily life. In addition, the research efforts focusing on a broad description of subjects' behavior also fed into the early development of EMA, which included but not limited to the time-budget research (Szalai, 1966) that used ethnographic method to describe individuals' allocation of time, and the Roger

Barker's Behavior Setting Survey. The latter project was acknowledge as the earliest efforts researchers made to incorporate contextual factors (e.g., physical context, social context) into methodologies for studying human behaviors (J. S. Barker, 2016; R. G. Barker, 1963; Georgiou, Carspecken, & Willems, 1996).

The development of modern EMA benefitted most from the development of "experience sampling method" (ESM; Csikszentmihalyi & Larson, 2014) and ambulatory monitoring (Hoppmann & Riediger, 2009). In psychology, ESM has been best known as a method for randomly sampling individuals' experiences and has been initially used as a signal-based data recording method that use beepers to prompt people at random times to report their internal experiences (e.g., thoughts, feelings) on dairy forms. Unlike ESM, ambulatory monitoring often paid little attention to sampling, because it typically was designed for continuous or near-continuous recording via wearable monitors (e.g., actigraph or heart-rate monitor) rather than relying on self-report data. Typically, the ESM focused on randomly sampling of subjective states, whereas self-monitoring focused on the occurrence of behaviors, and ambulatory monitoring focused on continuous recording of physiological parameters. Despite of these diverse traditions and distinct assessment targets, these methods have been included in the family of EMA because of their commonalties in goals, concerns, and approaches. Therefore, EMA has been used as an umbrella term to refer to a broader methodology that draws from a number of relevant approaches mentioned above to obtain repeated measures over time with emphasis on real-time (or near real-time) assessment under the real-life environments (Cain, Depp, & Jeste, 2009).

2.3.2 The key methodological aspects of EMA-based research

In this section, we gave an introductory overview of the key methodological features of EMA-based research. Unlike studies using retrospective survey methods, the most fundamental designing issue of EMA is about whether the moments or periods assessed could be representative of the person's real-life experiences. Specific to a research that aimed to examine within-person variations in everyday social and affective experiences, the assessments can be considered as a sample of phenomenon (the person's social behavior, event, and affective experiences). To obtain such a sample of behaviors and experiences, there are four major methodological aspects that researchers need to make decisions on when designing an EMA protocol.

2.3.2.1 Sampling Scheme

The scheduling of assessments should be driven primarily by the nature of the target experience or behavior as well as by the research question (Trull & Ebner-Priemer, 2009). There are two main types of sampling schemes: time-based sampling scheme and event-based sampling scheme. Time-based sampling scheme aims to characterize experiences inclusively and broadly without a pre-defined focus on particular discrete events, which can be divided into two sub-types. The first one is completely random time-sampling, which means that researchers randomly send a certain number of signals to the participants within a specified time of day (such as 8:00 am to 10:00 pm), so the interval between any two signals could be variable to ensure a representative sampling of participant's experiences (Shiffman et al., 2008). The second one is period-based time sampling that the researcher first divided the time of day into several large time periods (morning, afternoon, night) and send a signal to the participant within each period, or divided the time of day into more short periods (for example, in 15 minutes), and randomly selected some time periods to send signals. Under the period-based time sampling, the interval between any two signals could be either fixed or variable. This approach ensures that the assessment is able to extract the experience evenly throughout the day and that the assessment is included for each time period. The other major category is event-based sampling, which is designed to sample the discrete events in real time and ask the participants to report immediately after the target event occurs. For example, the subject is required to report after each cigarette is smoked. There were also studies combined the use of the event-based sampling and the period-based sampling (Stone & Broderick, 2007).

2.3.2.2 Study duration

Regarding the decision on the study duration, researchers need to consider at least two aspects. The first is the difference between working days and weekends. Many of the research questions, such as time allocation, leisure activities, work stress may vary significantly between workdays and weekends. Therefore, the researcher must include both working days and weekends in the number of research days to ensure that the obtained assessments have a representation of real-life experiences. The second point is the minimum number of observations the study required for testing relations between specific variables. This is especially true because most EMA studies focus on differences within individuals. For example, pervious research on everyday affective states have indicated that a minimum of 30 observations from the same person was needed for revealing significant variations in affective states. In order to achieve such a minimum number of observations, researchers could either enhance the study duration or the intensity of assessment. In addition, it is obvious that the study duration should be longer if the researcher was interested in the week-to-week variations rather than in the day-to-day variations.

2.3.2.3 Assessment intensity

The assessment intensity was primarily determined by the goal of the study and the knowledge about the target behavior or state. Importantly, researchers need to consider how rapidly the targeted behavior or states was expected to change during a certain period of time. Also, it is important that the assessment intensity should not be too high to avoid the boredom of the participants, which may further lead to a decrease in the response rate and the quality of the responses. In addition, researchers sometimes need a balance between the study duration and assessment intensity to obtain enough assessments required for testing specific hypotheses.

2.3.2.4 Signaling device

In terms of the data collection tools, EMA studies could be accomplished via a range of devices or platforms, including the most traditional paper-and-pencil diaries (often in a combined used of beepers), programmable watches, palmtop computers, automated telephone survey software, and smartphones. Early research usually relied on a pager or direct phone calls to ask the participants to response to required assessment or task. During the last two decades, technological advances have enabled more researchers to adopt the computerized methods in which data was collected by mobile devices (Bolger, Davis, & Rafaeli, 2003). Mobile-based EMA has largely solved the back-filling problem of the paper-and-pencil procedures through precisely controlled timing and tracking compliance (Granholm, Loh, &

Swendsen, 2007; Hufford, 2007). Also, researches could obtain a number of additional information regarding the compliance, for example, the exact time the participants started to response to the signal and how long it takes for each participant to complete the data entry or other tasks. By doing so, a delayed response could be identified, and if necessary, controlled for the specific analyses. Another advantage of mobile-based EMA is that it could flexibly accommodate the participants' daily life schedule. For example, participants are allowed to delay or even temporarily turn off the program when they do not want to be interrupted by the prompts (e.g., at a meeting, taking with the supervisor, in church). For more advantages of using mobile technology in EMA study, see Barrett and Barrett (2001) and Shiffman (2007) for more detailed discussions.

2.3.3 The feasibility of mobile-based EMA among aging population

As a data collection method, EMA holds a number of advantages (e.g., enhanced ecological validity; reduced recall bias; repeated assessments addressing the within-person relations) that enables its application to the study of a wide array of phenomena. However, there was only a very small number of studies that were conducted in older population to date. Despite of the largely increased availability and affordability of mobile technologies for the design of more sophisticated EMA protocols, the use of technology in these studies was still very limited, such as using an electronic device (e.g., beeper) to send signals to remind the older person to complete the paper-based survey (Cain et al., 2009). This reluctance might be attributable to the concerns about the feasibility of EMA for collecting data from older people, especially regarding older people's attitudes toward and capacity with the use of mobile technology. However, it has been reported that smartphone ownership among seniors continues to climb across countries and regions, suggesting that the ‘digital gap’ between young and older generations is rapidly shrinking (Anderson & Perrin, 2017; McMurtrey, McGaughey, & Downey, 2008; Poushter, 2016). Importantly, recent evidences indicated that older adults may actually prefer to work with the mobile-based EMA protocols over traditional paper-based EMA protocols (Shiffman et al., 2008) because of its greater ease of use through the multi-function (signal transmission and date entry), automatic administration of questions, and a great potential of personalization that can reduce the interference with respondents' normal lives (Rullier et al., 2014).

Although there has been a number of studies describing the feasibility and usability of mobile-based EMA across different populations, little information has been provided about the key designing and implementing issues for the success of a mobile-based EMA protocols among older population. Few of these studies have provided information about the full compliance rates among older participants (e.g., whether compliance rates change over time throughout the day, whether gender influence the compliance rate), and little is known about how to optimize user-based options, how to enhance older participants’ confidence in working on the EMA procedures that involve the use of mobile technology. Two exceptional studies (Fritz et al., 2017; Ramsey et al., 2016) evaluated the usability of smartphone-based EMA protocols in older Americans and reported satisfactory compliance. However, we do not know whether similar EMA approaches would also be feasible and acceptable when being applied to collect data from older Chinese. There is evidence suggesting that unlike the progress in North America, the digital gap in developing countries such as China is widening and deepening (Chen & Wellman, 2004; Pick

& Azari, 2008). Accordingly, researchers may encounter unique and different challenges when implementing sophisticated EMA protocols with older Chinese, who were relatively disadvantaged regarding the proficiency with mobile technology due to lower education and socioeconomic status. In order to maximize older people's adherence to the EMA, researchers also need to consider the issues of user privacy and trust, which may have different manifestations in different cultural contexts and significantly influence the success of the EMA implementation well as the quality of data researchers finally obtained . Overall, it is timely and important to provide reports about the feasibility of mobile-based EMA with a sample of older Chinese.

2.4 Summary and research gaps

This review has organized the evidence to date for a relationship between social interactions and affective states. It has been well-established that, one the between-person basis, being socially connected was associated with the betterment of affective well-being, and the strength of such association might vary according to "who is the interacting partner" (the uniqueness of the partner in terms of relationship type, familiarity, closeness). Beyond the structural (quantifiable) aspects of social ties, how people think about their surrounding interpersonal environment has been suggested as important predictors of short-term fluctuations in affective states.

Although great variations existed across studies in the assessment of people's perceptions of social interactions, there is agreement that individuals reported higher valence when interacting with familiar people, when they reported greater pleasantness during the interaction, or when they perceived more communal behaviors of their partners (Boehringer et al., 2010; Le & Agnew, 2003; Mehl et al., 2010). In contrast, elevated negative affect and arousal were reported by individuals when engaging in interactions characterized by negative features (e.g., disagreement, rejection) or when they perceived more agentic behaviors of themselves or from others (Luong & Charles, 2014; Thorsteinsson & James, 1999; Wrzus & Mehl, 2015).

In addition, many studies in this review examined the role of age and psychological traits in explaining the variability in affective states in relation to daily social interactions. Age, neuroticism, attachment styles, and loneliness appeared to be moderators exerting significant effects in the relationship between social interactions and affective experiences. These findings provided more detailed views about how the how between-person differences (traits) might operate to influence everyday dynamics of interpersonal experiences and affective states.

Despite these studies reviewed above enhanced our understanding of the interplay of everyday social experiences and affective states, we identified several important research gaps that needed to be addressed by subsequent studies. These research gaps included (1) few of the existing studies have properly disaggregated the within-person effects from the between-person effects of social interactions on affective states; (2) it remains unclear the bidirectionality of the relation between social interactions and affective states; and (3) little information has been provided about the feasibility of using mobile-based EMA to collect data on social and affective experiences from older people.

2.4.1 Separating within- and between-person effects of social interactions on affective states

So far, a large amount of research has examined the affective implications of social experiences from the between-person perspective. Their findings tended to conclude that on average, individuals who are more actively engaged in social environment enjoyed a higher level of emotional well-being. Even if we temporarily put aside the great variations in the established between-person relationship (which would suggest that this conclusion was too simplistic), knowing that socially connected people tended to be happier than those isolated does not allow an assumption that a person would be happier than typical when he or she interacts with others. Addressing the latter research question requires a within-person approach, which asks fundamentally different questions from the between-person approach, and accordingly, requires different data structure and analysis strategy to address.

Recently, there has been an increasing attention in many fields of psychology research paid to the within-person variations in affective states and their daily life correlates such as social interactions. It is important because affect is not a static phenomenon but may be altered across short-time scales (e.g., hours) when we go through different social interactions in a given day. Although we identified a handful of studies on the within-person covariation of social interactions and affective experiences, many of them did not properly disaggregate the between-person effects from the within-person effects of social interactions on affective experiences.

2.4.2 Testing bidirectional associations between social interactions and affective states

Most EMA studies in this review focused on the concurrent association between social interactions and affective states, and little efforts have been made to examine the temporal and lead-lag associations. Since most of the existing findings were interpreted to suggest that social interactions had impacts on affect, it remains unclear the possible bidirectionality of the relations between social interactions and affective states. It is plausible that different affective states would predict people's experiences of social interactions in the near future (e.g., does negative mood experienced at prior time point lead us to rate subsequent interactions as more negative). To test the temporal lagged associations between social and affective experiences, a denser measurement schedule should be used to provide more moments throughout the day. Then researchers could test lagged models to see how a study variable of interest assessed at prior occasion would predict another variable at a subsequent occasion. Such lagged relations would allow researchers to make stronger inferences about person-level causality (Brannon, Cushing, Crick, & Mitchell, 2016; Smyth, Juth, Ma, & Sliwinski, 2017).

2.4.3 Developing mobile-based EMA protocol for collecting data from older people

The mobile-based EMA protocols have been shown with a variety of benefits for enhancing the usability and validity of the EMA research, however, few of the existing studies provided the needed information for the success of designing and implementing a mobile-based EMA protocol with older population, for example, how to optimize user-based options, how to enhance older participants' confidence in working on the EMA procedures that involve the use of mobile technology. A key methodological issue remains unclear that whether mobile-based EMA protocol, which was typically more sophisticated than traditional paper-based EMA protocols, might be feasible for collecting date from older population. Although two exceptional studies (Fritz et al., 2017; Ramsey et al., 2016) evaluated the usability of smartphone-based EMA protocols in older Americans and reported satisfactory compliance, it remains unclear whether similar EMA approaches would also be feasible in different cultural context such as urban China and whether older Chinese would accept this research method. Furthermore, little was known about the unique challenges that researchers may encounter when implementing mobile-based EMA research with Chinese older adults. Therefore, it is timely and important to provide reports about how to develop a mobile-based EMA among older Chinese regarding the usability and feasibility of the protocol.

Chapter 3 Theoretical framework

The theoretical perspectives on the multi-dimensional nature of social relationships will be discussed first, followed by a brief overview of the circumplex model of affect. Drawing from these theories, we illustrated the role of the quantitative and qualitative aspects of social interactions for the long-term interindividual differences in affective experiences, as well as the short-term variations in momentary affective experiences. Lastly, we formulated hypotheses at the between-person level about how individuals differ from others in affective well-being in relation to their overall level of social interactions, as well as hypotheses at the within-person level about how the same person's affect may change as they went through specific social encounters that were characterized by differences in the occurrence and subjective satisfaction.

3.1 Theoretical perspectives on the multi-dimensional nature of social relationships

3.1.1 The social convoy model: an integration of relationship structure and function

The social convoy model (Antonucci & Akiyama, 1987, 1995) was rooted in the perspective of life span and life course history to understand the process of social relationship development. The core of the model is to illustrate the multidimensional nature of social relationships and their implications for health and well-being statues. Typically, the examination of social relationships involves both quantitative (objective) and qualitative (subjective) characteristics: the quantitative features (often referred to structural aspects) were typically measured in terms of the size of social network, the frequency of social interactions, and the composition of different relationship types (e.g., kin, close friends). The qualitative aspects of relationships include affect, perceived or received support, companionship, affirmation exchanges, and strain in some cases. According to the convoy model, both the quantitative and qualitative aspects of convoys are shaped by a number of personal determinants including factors such as age and gender (Kira S Birditt & Fingerman, 2003), as well as situational factors such as norms and values, marital status, role demands, resources, social position, ability, and positive and negative life experiences (Antonucci, Langfahl, & Akiyama, 2004; Litwin & Shiovitz-Ezra, 2011a).

Importantly, both the quantifiable (objective) and qualitative (subjective) aspects of social convoys could exert significant impacts on health and well-being. For example, empirical researches utilized an individual -centered approach (instead of a variable-centered approach) to integrate a number of quantitative features of social networks (the size of networks, the relationship types, and contact frequency) (Fiori et al., 2006; Fiori et al., 2007; Litwin & Shiovitz-Ezra, 2011a) They had identified robust social network patterns (e.g., family-oriented, friend-intensive), and linked these patterns to individual differences in mental well-being, morale, and loneliness over time (Kira S Birditt & Antonucci, 2007; Kira S Birditt, Jackey, & Antonucci, 2009; Fiori et al., 2006; Fiori et al., 2007). Importantly, a growing number of evidences suggested that the quantity of social convoy might have particular relevance in enhancing emotional well-being in late adulthood (Mejia & Hooker, 2014). For example, researchers found that relative to quantitative network characteristics, subjective appraisals of the network such were stronger predictors of the risk of social isolation, and certain qualitative features of convoy such as perceived social support has been associated with fewer depressive symptoms among older adults (DuPertuis et al., 2001; F. R. Lang & Carstensen, 1994). In brief, the social convoy model elaborated the predictive utility of both the quantitative and qualitative features of the social networks for health and well-being consequences (Litwin & Shiovitz-Ezra, 2011a). However, this theoretical perspective provided little insights into the underlying mechanisms through which involvement in social convoys may influence health and well-being statuses.

3.1.2 A cube model of personal relationships: an integration of relationship structure, processes, and outcomes

Another theoretical model elaborating the multidimensional nature of social relationships is a cube model introduced by Fingerman and Lang (2004). This model is rooted from a life-span developmental perspective to understand how the structures, processes, and outcomes of social relationships are interwoven with human development. Specifically, the *structure* of social relationships was constituted by different social partners, for example, family and friends, close and peripheral ties, short-term and long-term relationships. The term of *processes* refers to the mechanisms underlying of and arising from social relationships, including the cognitive and emotional capacities of individuals, (dis)satisfaction with relationship, and motivations to connect with others. The *outcomes* typically include the (perceived or actual) benefits and costs of an individual's social relationships, including supports or demands, health outcomes, and affective well-being outcomes. The cube model devoted particular attention to the associations among the three dimensions and how they might change in response to the changing life events and circumstances across the adult life span (Fingerman et al., 2013).

Similar with the social convoy model, the cube model also considered how the structure of social networks shifts across life span (the types of social partners people encounter, the nature of specific relationships) and how the continuity and changes in relational structure might influence outcomes such as health and well-being. For example, since the loss of intimate ties (such as a spouse) becomes more common in later adulthood, the contribution of friendships to well-being outcomes might become increasingly important as people age (Fingerman & Baker, 2006).

Unlike the social convoy model, the cube model also directly focuses on how the processes underlying social relationships may influence the extent to which people benefit from their relationships with others (Fingerman et al., 2013). Particularly for the contribution of social ties to affective well-being outcomes, a growing number of researches pointed out that not only the long-term structure of social relationships (often measured in a global time frame) matters for affective well-being, but the immediate feelings and appraisals emergent from ongoing social interactions significantly contributed to individuals' affective experiences (R. Larson et al., 1986; Vittengl & Holt, 1998b; Vogel et al., 2017). In summary, the cube model of social relationships provided important theoretical foundation for linking the processes of social relationships to affective well-being outcomes (H. Chui et al., 2014). Our study built on the cube model and the social convoy theory to include both the quantifiable and qualitative features of the social networks, with a particular focus on the process dimension of the social interaction, that is, the immediate feelings and appraisals emergent from ongoing social interactions. In specific, the quantitative feature of social interactions was examined by the occurrence of interactions with different social partners across the day. The process dimension was examined by the participant's momentary satisfaction with the specified interaction. For the outcome of the social interaction, we examined momentary affective states that were emergent from day-to-day social interactions.

3.2 The circumplex model of affect

The circumplex model of affect is the most widely studied framework that takes a dimensional approach to capture the underlying structure of affective experience (Russell, 1980; Schlosberg, 1952). This model postulates that all affective states are generated from two fundamental neurophysiological systems, the valence dimension (a displeasure– pleasure continuum) on the one hand and the arousal dimension on the other side (a deactivated or sleepy – activated or energetic continuum). Each affective state can be conceptualized as varying degrees of, or a linear combination of both the valence and arousal dimensions (Posner et al., 2005). For instance, happy as one of the most widely studied affective state, could be considered the product of a strong activation in the neurophysiological systems related with positive valence (the pleasure end) coupling with a moderate activation related with arousal (a value between the sleepy and energetic). Sad is an affective state arising from a strong activation in the neural systems associated with negative valence (the displeasure end) coupling with a moderate activation associated with arousal (a value between the sleepy and energetic). Accordingly, all the other affective states (e.g., stressed, bored, depressed, excited, relaxed) are cosndiered as arising out of patterns of activation within the same two neural systems, and two specific affective states could be distinguished by the extent of activation in these two systems.

This circumplex model of affect has gained substantial empirical support from psychometric studies since its introduction. Using principal-components or factor analyses of self-reported affect and multidimensional-scaling analyses of similarity judgments of affective states, a substantial number of psychometric assessments have consistently reported the emergence of valence and arousal as the two underlying dimensions of affect when an individual was asked to label his or her own affective states or that of others (Barrett & Russell, 1999; Västfjäll, Friman, Gärling, & Kleiner, 2002). These reports have been replicated over a variety of time frames and response formats (Plutchik & Conte, 1997). Also, a number of researches using similarity ratings of facial expressions and emotion-denoting words to suggested a two-dimension model of affective states (P. J. Lang, Greenwald, Bradley, & Hamm, 1993; Meng & Bianchi-Berthouze, 2014; Remington, Fabrigar, & Visser, 2000). These findings have been generalized to different sample populations as well as to cross-cultural samples (Huang, 2003; Ong & Bergeman, 2004; Watson et al., 1988; Wilhelm & Schoebi, 2007).

This circumplex model has also been used as the most important measurement paradigms to assess the intra-individual variability in affect (Barrett, 2005; Watson, 1988), which referred to the short-term fluctuations that occurred across time or situation, including both the amount of variation as well as the more time-structured dynamics of affective experience (Hülür et al., 2015; Riediger & Rauers, 2014; Rocke, Li, & Smith, 2009). Such fluctuations in affective states have been studied either as an independent variable or as an outcome, and the majority of the methodological approaches to such fluctuations are based on repeated measurement designs, in which individuals are asked to rate their momentary affective states multiple times per hour, day, week, month, or year (Solhan, Trull, Jahng, & Wood, 2009; Wrzus & Mehl, 2015). In keeping with this research, the current study examined individuals' affective states that occurred across hours and days, of which we distinguished between high-arousal positive affect (e.g., excited) and low-arousal positive affect (e.g., relaxed), as well as between high-arousal negative affect (e.g., angry) and low-arousal negative affect (e.g., sad). Drawing from the circumplex model of affect and theories on the multidimensional nature of social relationships, we formulate hypotheses concerning whether and how individuals' affective states might differ or change in relation to their overall level of social interactions (between-person effects) and as they went through specific social encounters that were characterized by differences in the occurrence and subjective satisfaction (within-person effects).

3.3 Conceptual framework of the study

The conceptual framework (see Figure 3.1 and 3.2 for a visualized overview) for this study focused on two key constructs: social interactions and affective states. The operationalization of social interaction was built on theories about the multidimensional nature of social relationships, which specified two aspects of social interactions: (1) the quantitative aspect was examined by the occurrence of social interaction, which was defined by a face-to-face interaction with other people that lasted for 5 – 10 minutes or longer; (2) the qualitative aspect was examined by the subject's satisfaction with a reported interaction. The operationalization of affective states was based on the circumplex model of affect, which considered each affective state as a linear combination of two independent dimensions: the valence dimension (a negative-positive continuum) and the arousal dimension (a deactivated – activated continuum). Low-valence combined with low-arousal was considered low-arousal negative affect (LNA), low-valence combined with high-arousal was considered high-arousal negative affect (HNA); while high-valence combined with low-arousal was considered low-arousal positive affect (LPA), high-valence combined with high-arousal was considered high-arousal positive affect (HPA).

Second, this framework has the central focus on the separation of the between-person and within-person relations of social interactions and affective states. Conceptually, this framework was proposed to address two sets of fundamentally different research questions. The first set of research questions and corresponding hypotheses (Hypotheses 1 and 2) concerned about how individuals' average level of affect may differ in relation to their average level of social interactions in terms of the average level of occurrence and satisfaction (between-person relations). The second set of research questions and corresponding hypotheses (Hypotheses 3 and 4) concerned about how individuals' momentary level may change as they went through specific social interactions in terms of the momentary level of occurrence and satisfaction (within-person effects).

Third, we make reference to empirical evidences to extend this framework into an examination of the bi-directionality of the temporal relationship between social interactions and affective states. Specifically, we examined whether aspects of social interactions at a prior occasion would lead to impacts on affective experiences (Hypothesis 5 and 6), as well as whether the level of affective states would predict how people experience their future interactions (Hypothesis 7).

Specifically, the occurrence of social interactions was considered a situational factor that may influence affective states, and we only expected prior affect to predict future appraisals of interactions rather than future occurrence of interactions.

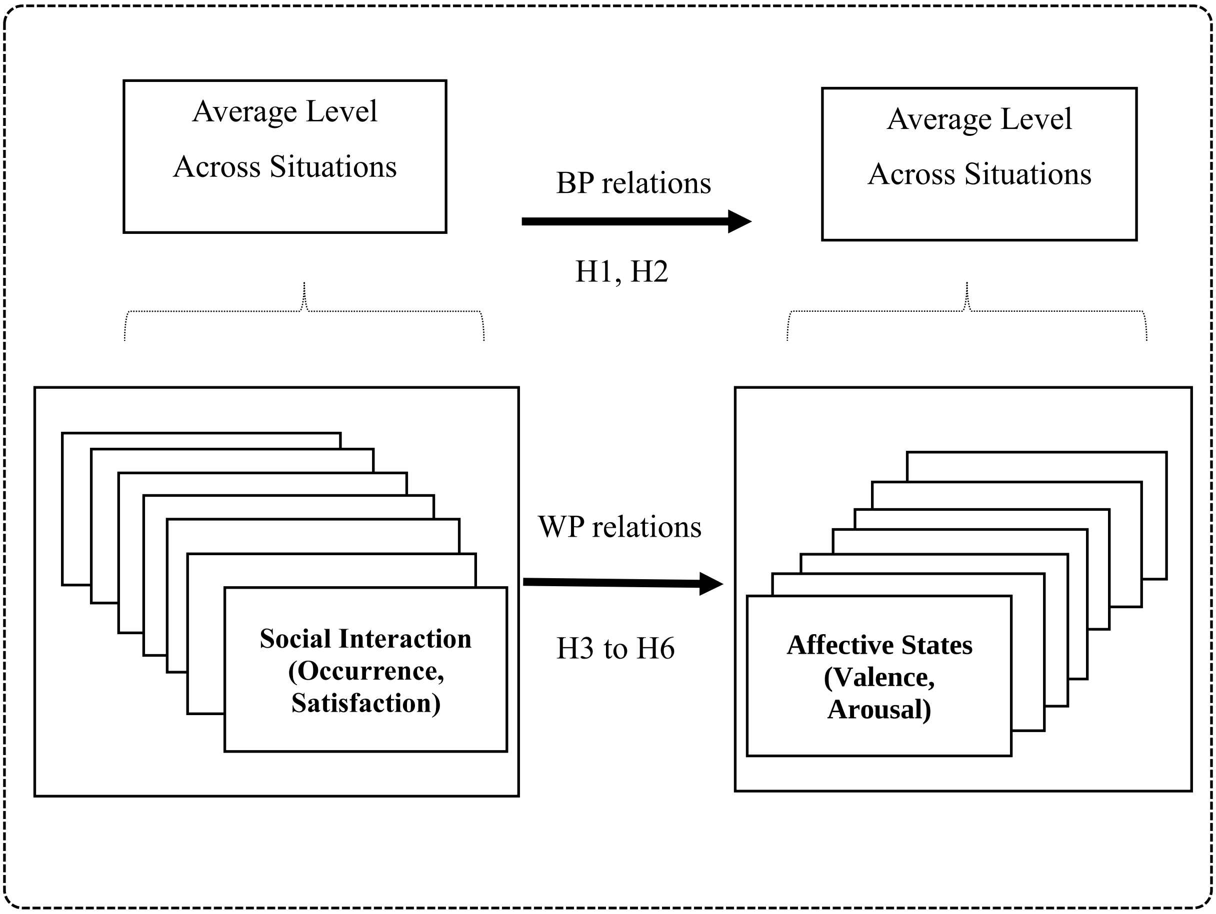

Figure 3.1 The framework showing the between-person (BP) and within-person (WP) relations of social interactions and affective states

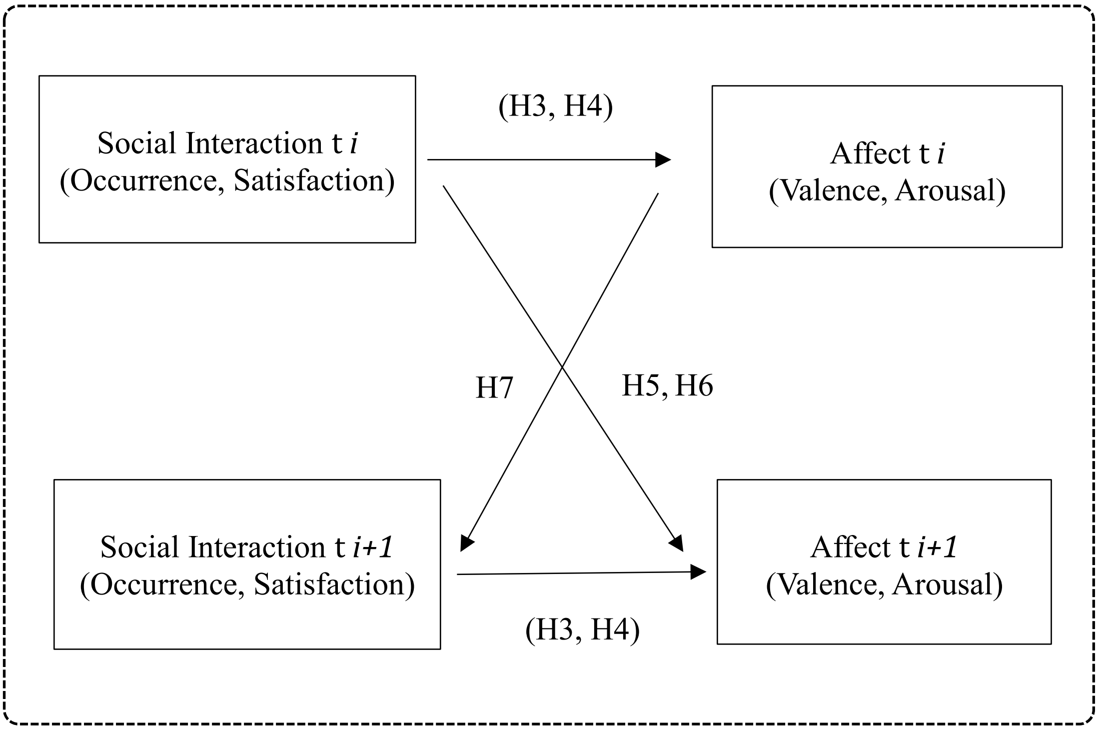

Figure 3.2 The framework showing the hypotheses about the lagged relations of everyday social interactions and affective states

3.4 Research objectives and hypothesis

The current research proposed four major research objectives. While research objective 1 is proposed for the methodological purpose, the other three objectives were for specific hypothesis testing.

Research Objective 1

To describe how to design and implement a mobile-based EMA protocol for collecting psychosocial data, and to evaluate the feasibility and acceptability of this developed protocol among a sample of older Chinese.

Research Objective 2

To test the overall between-person associations between the occurrence of and the satisfaction with social interactions with the average level of momentary positive and negative affect.

Hypothesis 1

H1a: Individuals who reported more social interactions on average would have higher level of high-arousal positive affect (HPA) than those who reported less interactions.

H1b: Individuals who reported more social interactions on average would have higher level of low-arousal positive affect (LPA) than those who reported less interactions.

H1c: Individuals who reported more social interactions on average would have lower level of high-arousal negative affect (HNA) than those who reported less interactions.

H1d: Individuals who reported more social interactions on average would have lower level of low-arousal negative affect (LNA) than those who reported less interactions.

Hypothesis 2

H2a: Individuals who reported more satisfactory social interactions on average would have higher level of HPA than those who reported less satisfactory interactions.

H2b: Individuals who reported more satisfactory social interactions on average would have higher level of LPA than those who reported less satisfactory interactions.

H2c: Individuals who reported more satisfactory social interactions on average would have lower level of HNA than those who reported less satisfactory interactions.

H2d: Individuals who reported more satisfactory social interactions on average would have lower level of LNA than those who reported less satisfactory interactions.

Research Objective 3

To test the concurrent and lagged within-person effects of the short-term variations in occurrence of and satisfaction with social interactions on the short-term variations in the level of positive and negative affective states.

Hypothesis 3

H3a: When individuals engaged in an interaction, they reported higher level of HPA than when they did not.

H3a: When individuals engaged in an interaction, they reported higher level of LPA than when they did not.

H3a: When individuals engaged in an interaction, they reported lower level of HNA than when they did not.

H3a: When individuals engaged in an interaction, they reported lower level of LNA than when they did not.

Hypothesis 4

H4a: When individuals appraised a social interaction as more satisfactory than that is typical for them, they reported higher level of HPA.

H4b: When individuals appraised a social interaction as more satisfactory than that is typical for them, they reported higher level of LPA.

H4c: When individuals appraised a social interaction as more satisfactory than that is typical for them, they reported lower level of HNA.

H4d: When individuals appraised a social interaction as more satisfactory than that is typical for them, they reported lower level of LNA.

Hypothesis 5

H5a: The occurrence of social interaction was followed by higher level of HPA.

H5b: The occurrence of social interaction was followed by higher level of LPA.

H5c: The occurrence of social interaction was followed by lower level of HNA.

H5d: The occurrence of social interaction was followed by lower level of LNA.

Hypothesis 6

H6a: Higher satisfaction with social interaction was followed by higher level of HPA.

H6b: Higher satisfaction with social interaction was followed by higher level of LPA.

H6c: Higher satisfaction with social interaction was followed by lower level of HNA.

H6d: Higher satisfaction with social interaction was followed by lower level of LNA.

Research Objective 4

To test the lagged within-person associations between momentary affective states at prior time point (t-1) and the subsequent level of satisfaction with social interactions.

Hypothesis 7

H7a: Higher level of HPA was followed by higher satisfaction with social interaction.

H7b: Higher level of LPA was followed by higher satisfaction with social interaction.

H7c: Higher level of HNA was followed by lower satisfaction with social interaction.

H7d: Higher level of LNA was followed by lower satisfaction with social interaction.

Chapter 4 Methodology

4.1 The selection of the case city

We selected Changsha as the site for data collection. There are six urban districts in Changsha (Kaifu, Furong, Yuelu, Tianxin, Wangcheng, and Yuhua), in addition to two counties and one county-level city. In 2012, Changsha's gross domestic product (GDP) ranks seventh of a list of provincial capital cities in mainland China (639.99 billion yuan) (Zhang Jianfei, 2011; Hunan Provincial Bureau of Statistics, 2013). The GDP of Changsha (Wang, 2013). According to the data of the sixth national census that was conducted in 2010, the population of Changsha is 7,044,118, with an almost identical distribution of gender (China State Council Information Office, 2011). It was in 2011 that Changsha begun to become an aging society. The aged population (defined as aged 60 and over) accounts for around 16% of the city's total population; the average life expectancy at birth of residents in Changsha has been over 76 years by the end of 2010 (Li, 2011).

We selected Changsha as the site city for the following reasons. First of all, Changsha is a medium-level city in China (the capital city of Hunan Province in central and southern China). Unlike those super bid cities like Shanghai, Beijing and Shenzhen, which have been equipped with strong financial strength and social welfare preferential policies, Changsha is more like one of the ordinary cities in China that are facing the challenge of a rapidly growing number of aging populations. Secondly, it was at the beginning of the 21st century that Changsha entered an aging society; Changsha’s population aging is still on the rise. Between 2005 and 2010, the growth rate of aging population was 1.2%. In addition, Changsha can represent some cities located in central-south part of China to a certain extent. The field study of this city can help to reveal the overall development model of many other cities in China that were comparable to Changsha to a certain extent (Wang, 2012). Last but not least, in order to facilitate field research (especially considering the labor-intensive nature of the EMA method), the social networking and connections of researchers in the city can significantly help to overcome the barriers and difficulty of conducting intensive and in-depth assessments of older people. This factor has been acknowledged as an extremely important determinant of the success of the primary data collection in mainland China.

4.2 Participants

Given the purpose of obtaining a sample to be reflective of the distribution of gender and age-groups within the city's late-middle aged and older population, this study used the purposeful quota sampling method to recruit participants at the initial stage. First, three out of six districts in Changsha city were selected (Kaifu district, Wangcheng district, Tianxin district), followed by a random selection of one street office from each district. The research team initiated the contact with the administrative staff from each selected street office and got their support in providing the names of the persons who worked in the affiliated neighborhood committees or the persons who has strong connections to the local neighborhoods. Second, the research team approached a number of persons appeared at the name list. Within each street, one suitable person was employed as the part-time research coordinator responsible for connecting the research team and the local communities. Third, the three research coordinators started to contact the potentially eligible participants living in their responsible communities and gave brief introduction about the purpose and procedures of the present study. An eligibility screening was conducted for those persons who demonstrated interest in involving in this study. If the person met all the eligibility criteria, the research coordinator would help him or her to arrange a meeting with the primary investigator (either in the researcher's office or the participant's home). During this appointment, the researcher introduced the study procedures, the ethical issues, the content of the study consent forms in details. After the clarification, the potential participants made the decision about whether he or she would participate in this study.

At the final stage of participant recruitment, we added snowball sampling method into use following successfully enrolling about 50% of participants. This is because previous studies have shown that the incorporation of snowball sampling methods can effectively recruit participants from hard-to-reach subgroups (Liu & Lou, 2018; Sadler, Lee, Lim, & Fullerton, 2010). In our case, snowball sampling was used as a supplement to enhance participation of the subgroup of male participants aged 50-60, who have been rarely recruited in prior EMA studies. Specifically, we encouraged enrolled participants to recommend potential participants and help us to initiate the contact with the person. If eligible, we also followed the same procedure described above to enroll this person.

The eligibility screening was conducted based on the following inclusionary criteria: (1) the age of the participant was between 50 and 70, (2) the participant lived independently in the community (e.g., here 'independence' was defined by the absence of need to receive help in activities of daily life from another person); (3) the participant has sufficient hearing and vision capacity to answer questions displayed in the screen of mobile applications; (4) the participant had a sufficient cognitive ability (in our study, defined by a score of 26 or above on the Montreal Cognitive Assessment (Beijing versio) after adjusting his or her education attainment (Lu et al., 2011); (5) the participant did not have a history of any cognitive and mental disorder; (6) the participant did not have a history/diagnoses of cardiovascular diseases and did not take any relevant medicine currently.

Overall, the screening was conducted for 167 potential participants, of which 95 individuals were identified as eligible for participation in the current study. The excluded 72 individuals reported a mean age of 64.03 (SD = 10.57), 48 being female, 13 reported needing or receiving help from others in activities of daily life, 7 reported a MoCA score under 26, 8 reported having insufficient ability (hearing or vision) for completing the EMA procedures using the mobile application, and 39 reported having a history/diagnosis of cardiovascular disease and/or currently taking some medicine for treating/controlling this disease. Of the 95 individuals offered participation, a final sample of 78 participants were successfully enrolled (59 participants were from random selection and 19 from snowball method). The other eligible persons (n = 17) refused to participate due to the following reasons: being unavailable (e.g., travelling plan) during the proposed data collection period (n = 8), having demanding working or caregiving task (n = 3), perceiving it difficult to adhere to the procedures (n = 5), or refusing the physiological measurement procedure (n = 1). Following enrollment, there is no participant withdrew from the study. Each participant received a cash compensation (HK$ 500) once completed all the study procedures.

4.3 Study protocol design

4.3.1 Baseline interviewer-administrated geriatric assessment

Before the EMA data collection period, trained interviewers conducted the baseline self-administrated survey for each participant. Most of the interviewer-administered surveys were taking place at the apartment of the participant or in the corresponding community service center. The baseline assessment covered the following eight parts: (1) socio-demographics and family information, (2) physical functioning, (3) history of chronic conditions; (4) cognitive functioning; (5) social networks and social support; (6) health-related quality of life; (7) depression, and (8) health behaviors and lifestyle.

4.3.2 A five-step training session

Before entering the EMA study, we loaned each participant a smartphone with Android system, which has been installed with a research-developed EMA survey application. In order to maximize participants' adherence to the study procedure and data quality, we provided all the participants an in-person training session, either taking place in their home (n = 65) or in the office room of the trained research assistant's (RA) (n = 13). The training session was designed based on the existing research using the EMA approach to collect data from community older populations (Fritz et al., 2017). The entire training is designed to be divided into five phases to ensure that the elderly can gradually improve their confidence and comfort with working on the EMA application (Brannon et al., 2016; Kuntsche & Labhart, 2013; Mareva et al., 2016).

(1) In step one, the RA gave a verbal overview to introduce the basic function (turn

on and turn off) of the smartphone. We observed that for most participants who has the experiences using Android smartphones (n = 65), this verbal overview and demonstration took only 5 to 10 minutes. For participants who had never used a smartphone, a more detailed instruction was provided.

(2) After achieving the goal of step one, the RA started to give instructions on how to use the researcher-developed EMA survey application. A verbal overview of this applications was provided first, followed by an illustration of how to access the application (touch a specified icon to enter the interface and press the "Start" at the bottom of the interface to start answering questions), how to response to the questions in the screen, and how to exit the survey. This step took the participants 10-20 minutes for most participants. For participants who had never used a smartphone, a more detailed instruction that took additional 10-20 minutes was provided.

(3) In the third step, the RA encouraged the participants to show what they have learned from the former steps by accessing the application and responding to the required tasks independently. The RA provided additional explanations and instructions to help participants to solve the problems they encountered during the demonstration (e.g., how to return after touching other function keys).

(4) In step four, a take-home handbook that covered all the necessary information the participants need to know about the procedures was reviewed by the RA and distributed to each participant, which covered the use guidelines of the EMA application, the protocol to refer to, and troubleshooting help during the real data collection period.

(5) Before the real data collection, a final step is allowing each participant to have one or two practice days. On the practice day, the RA continuously monitored participants for difficulties in complying with the protocol through a web-based interface. Timely assistance was provided through phone call or home visit when problems occurred. We did not install the Subscriber Identity Model (SIM) card in the smartphones loaned to our participants. Participants carried their own phones for personal use (e.g., make phone calls) during the study period.

4.3.3 Procedures of the one-week EMA data collection

The EMA data collection phase lasted for one week (seven consecutive days) for each participant. For each assessment day, the 12-hours timeframe between 8 am and 8 pm are divided into six two-hour periods (8am - 10am; 10am-12am; .etc.), and the EMA application randomly sent a sound alarm to remind the participant to complete a brief survey during each of the six time periods. A voluntary assessment was scheduled between 8:00 pm to 10:00 pm, which means that the participants could initiate the data entry without receiving a sound alarm. The application was programmed to ensure that the prompts were scheduled with different random patterns for each day, with the minutes between two close-in-time prompts being greater than 30 and less than 150. In principle, we encouraged our participants to response once they received a prompt but a delay for a maximum of 15 min was allowed. The application recorded the start and completion time of each assessment. If the participant did not response to the prompt over 15 minutes, substitute prompts was automatically sent every 10 minutes for 2 times. A missing was recorded if all the three signals received no responses.

To maximize cooperation, we designed the content of the brief survey (displayed in the screen of the smartphone) to be short and easy to complete. A total number of twenty-seven questions were programmed into the EMA survey application. Several skip patterns were designed to reduce the number of questions posed at each assessment. The content of the EMA survey covered three general sections: (1) questions about the current context (the primary activity the subject is now performing, duration of the activity, current location, etc.), (2) characteristics of reported social interactions, and (3) affective states.For example, if the participant reported being alone at the moment, the participant skipped the questions about the occurrence and the quality of social interactions and went directly to the last section about his/her current affective states.

During the one-week data collection, the RA kept monitoring the data entry progress and providing continuously on-call help. Based on our platform design, participant's survey data was directly stored in the smartphone, and once the participants had access to Internet, it can be automatically saved into our database through the transforming from smartphone to the "cloud data storage system". During the data collection period, we found that a very few of participants are often able to connect to the Internet throughout the data collection week. Therefore, most of the participants' survey data have been manually transported into our database after the data collection week (e.g., the RA retrieved the phone loaned to the participants). See Figure 4.3 for an overview of the one-week EMA data collection procedure.

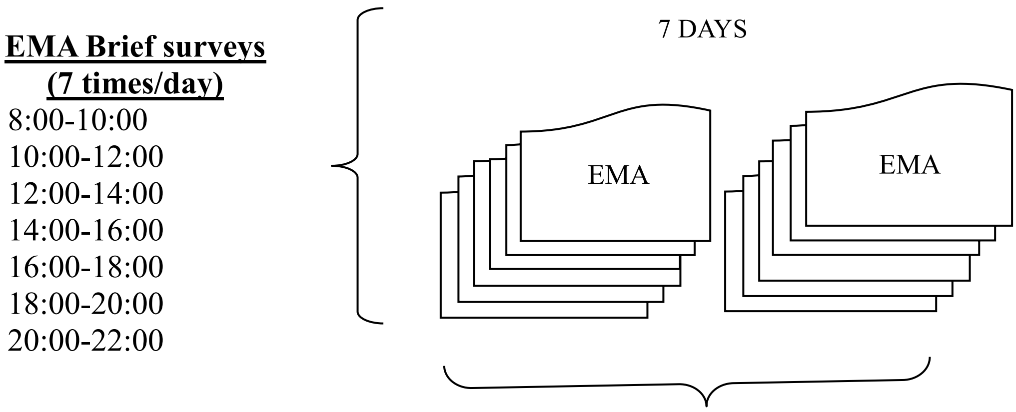

Figure 4.3 An illustration of the EMA research protocol

4.3.4 Follow-up interview

The RAs visited the participant's homes to conduct a semi-structured follow-up interview within two days after the EMA data collection week. Participants were invited to complete an evaluation form that included 14 items revised from prior studies on the feasibility and usability of EMA, as well as to share their experiences and their suggestions to improve the feasibility and usability of the EMA protocol (Granholm et al., 2008; Ramsey et al., 2016).

4.3.5 Modifications made to the pilot version

Before the actual data collection period, we conducted a pilot study that included a separate sample of 10 older volunteers living in the local community (mean age = 56.72, SD = 9.25; 5 women and 3 men) . We followed the procedure described earlier except that the duration of the trial study was shorter (range = 1 to 3 days, mean = 1.7 days, SD = 0.82). After the evaluation date, participants received interviews with their experience using the EMA protocol, during which a series of questions were discussed, such as the difficulties they encountered in completing the brief survey and ECG recording, and the adequacy of participant training. Generally, these participants reported that they have a good understanding of the protocol and how the application works after providing training. Most participants considered their experience positive, and some are even excited to get more familiar with mobile technology.

It is worth noting that most participants stated that they preferred more autonomy during the assessment period. In this regard, we invite them to elaborate on the relevant issues, and propose improvements to the EMA to provide more user-based elements. For example, two participants felt it was difficult to respond to the first alert between 8:00 am and 10:00 am because they often got up after 9:00 am. Based on these feedbacks, we allow our participants to turn off the alarm when needed and add an optional time interval (20:00 - 22:00) without an alarm to accommodate the different wake-up time patterns of the participants.

Another major criticism of the EMA interface is that when the prompt does not receive a response within 20 minutes, the participant is no longer allowed to access the answer page until the subsequent time interval begins. The two participants made it clear that it was stressful to respond to these intensive tips in a timely manner (within 20 minutes), one of them said "it's hard, I have to say that I don't like this appealing feeling. Many times I can't do this (respond to the prompt), just like I am anxious to pick up my grandson from kindergarten. Another person said: "I am a little nervous, I really want to do a good job, do more times, you know I feel very sorry when I miss a reminder. I want to delay for a while." Based on this feedback, the revised EMA keeps the answer page open at each time interval when the assessment has not completed. This means that participants can initiate an assessment before or after they receive a prompt in each time interval. In addition, we allow a participant to turn off the alarm for most of the time to avoid disturbing the two frail family elders who provide primary care.

After incorporating these changes, we interviewed 10 participants who were the first participants to start real data collection immediately after the warm-up day. These interviews did not produce further changes, and most participants were satisfied with the key changes we made to the EMA agreement. All participants used the same revision EMA interface during actual data collection.

4.4 Measurement

4.4.1 Baseline questionnaire measures

Baseline questionnaire measures were selected to cover a range of between-person level covariates of social interactions and affective experiences, covering the following seven parts. (1) Socio-demographics and family information included measures of age (measured in years), gender (female = 0, male = 1), education attainment (less than high school = 0, high school or more = 1), family income (less than RMB 3,000 = 0, RMB 3,000 or more = 1),retirement status (full-time employment = 0, partially retired = 1, fully retired = 2, unemployed = 3). (2) physical functioning included measures of mobility limitations (seven items mearing whether the participant having difficulties in walking short distances, climbing stairs, getting up from a chair, stooping or kneeling or crouching, etc.), difficulties in performing activities of daily living (ADLs, including eating, toileting, etc.) and instrumental activities of daily living (IADLs, including cooking, shopping, etc.), (3) history of chronic conditions included measures of the year diagnosed by a doctor with a number of chronic conditions (e.g., hypertension, diabetes, etc.); (4) cognitive functioning was measured by the Chinese version Montreal Cognitive Assessment (Lu et al., 2011); (5) social networks and social support included measures of the six-item Lubben Social Network Scale (LNSS-6) and measures of perceived social support (Wilcox, 2010); (6) health-related quality of life was measured by the Short-Form 12-Item Health Survey (SF-12) (Ware Jr, Kosinski, & Keller, 1996); (7) depression was measured by the Geriatric Depression Scale (GDS) (Chan, 1996; Lee, Chiu, Kowk, & Leung, 1993); and (8) heath behavior and lifestyle included measures of smoking during the past year, drinking during the past year, and questions about physical activity (e.g., The International Physical Activity Questionnarie, Lee, Macfarlane, Lam, & Stewart, 2011).

4.4.2 The within-day EMA measures

4.4.2.1 Questions about the Contextual information

At the beginning of each EMA brief survey, the participant was asked to report, at the current moment, the type of activity he or she was concurrently performing from a list of 18 activities, his or her location (0 = home, 1 = indoor, 2 = outdoor), the duration of the performed activity (0 = 1 to 5 minutes, 1 = 6 to 10 minutes; 2 = 11 to 20 minutes; 3 = 21 to 30 minutes; 4= 30-60 minutes; 5 = over 60 minutes); the physical level of performing this activity (0 = no physical activity, 1 = light; 2= = moderate; 3 = heavy); and the presence of others (with others versus alone). If the participant was not alone, he or she needed to continue to answer whether he or she was interacting with the presented person(s). We selected these questions to obtain immediate contextual information of the individuals based on the existing literature and the Diary of Ambulatory Behavioral States (DABS), for the original work, see Kamarck et al. (1998). For the type of the primary activity the participant was performing, we developed a list of pre-defined activities based on previous work on older adults' activity engagement (for example, the categorization developed by Maier and Klumb (2005) that distinguished regenerative activities and discretionary activities). These activities included: (1) eating, (2) personal care, (3) housework, (4) paid working, (5) caring for others, (6) volunteering, (7) informal helping, (8) reading/listing to radio, (9) watching TV, (10) using electronic devices; (11) active leisure, (12) exercise, (13) shopping, (14) locomotion, (15) medical activities, (16) social activities, (17) learning activities, and (18) other activities.

4.4.2.2 Questions about the features of reported social interactions

In the former part (contextual information), the last question was about whether the participant was interacting with the presented person. If the participant reported yes, he or she was further instructed to respond to the current section about features of social interactions. This section included the following questions: (1) the interacting partner's relationship with the participant (spouse, parent, children, grandchildren, other family members, friends or neighbors, coworkers, others); (2) the duration of the interaction (0 = 1 to 5 minutes, 1 = 6 to 10 minutes; 2 = 11 to 20 minutes; 3 = 21 to 30 minutes; 4= 30-60 minutes; 5 = over 60 minutes); (3) the initiator of the interaction (1 = the subject, 2 = the partner, 3 = not sure); and (4) the self-rated satisfaction with the interaction (*please indicate to what extent you felt satisfied with this reported interaction on* a scale ranged from 0 = not at all satisfied to 9 = totally satisfied). These measures were adapted from prior research on qualitative features of social interactions and interpersonal behaviors (Bernstein et al., 2017; Brondolo et al., 2003; Mejia & Hooker, 2014).

4.4.2.3 Momentary affect

We selected nine items to assess momentary affective states from the literature on the intra-individual variabilities in state-level affect. These included three items for HPA (*happy*, *excited, alert*), two items for LPA (*calm* and *quiet),* two items for HNA *(anxious*, *irritated*), and three items for LNA *(sad, sleepy, lonely).* These items were scaled from 0 = not at all to 4 = very much. We selected these items for three major reasons. First, prior research suggested that each of these items displayed considerable within-person variations that reflected the short-term changes in affective states. Second, the selection was theory-driven, because these items in total covered the two dimensions, valence and arousal of the affective circumplex model. In addition, these items have been widely used in ecological momentary assessment researches and showed satisfactory longitudinal reliability in the context of ecological momentary assessment design (intensive longitudinal design) (Shrout & Lane, 2012; Tsai, Knutson, & Fung, 2006). Third, these items were commonly used in EMA-based affect research on older population.

Previous studies have suggested that in the context of a EMA design, these items showed reasonable psychometric properties (Shrout & Lane, 2012), and the psychometric properties of these items have already been reported in Chinese context (e.g., internal consistency, reliability, construct validity). Specific to the current study, the internal consistency was less than optimal: Mean Cronbach's alpha across the within-person assessments ranged from 0.48 to 0.59 for affect subscales. There are three reasons for the less-than-optimal internal consistency: (1) the limited number of items for each sub-scale (2 or 3 items) for reducing participant burden; (2) the small sample size of individuals; (3) For each sub-scale, we selected a heterogeneous set of items (rather than homogeneous) in order to capture the wide range of valence and arousal dimension. In brief, the cross-sectional reliability of these items was less-than-optimal but acceptable in the context of EMA settings.

Importantly, our measurement of affect showed excellent longitudinal reliability. Four indices based on the generalizability theory (Cranford et al., 2006; Shrout & Lane, 2011) suggested the selected items were sensitive to detecting moment-to-moment changes in affect (RKF .99 for positive affect and .99 for negatvei affect; R1R .75 for positive affect and .78 for negative affect; RKR .98 for positive affect and .98 for negative affect; and RC .37 for positive affect and .36 for negative affect).

4.5 Analytic strategy

Descriptive analyses were used for describing the baseline characteristics and the response rates during the one-week EMA period for the whole sample. Between-group differences in terms of the baseline characteristics were compared for the participants selected through quota sampling (n = 59) with participants selected through snowball sampling (n =19). Multilevel modeling (MLM) approaches were used given that the collected EMA data has a hierarchical (nested) structure: here the observation level (multiple assessment per day) was the lower level of the hierarchy that was denoted Level 1, and the person-level was the higher level of the data structure denoted as Level 2. The major reason for using MLM is that, and

MLM was recommended for handling EMA data and was ideally suited for the current study because of the following major advantages: (1) lower-level coefficients (the observation-level) can be included in the model as dependent variables (allow us to our test the within-person variations); (2) missing values presented at variables measured at level 1 across participants not pose a serious problem to the estimation; (3) the absence of severe parameter bias that researchers may encounter when data were handled without considering the nested structure.

In specific, all of our models allowed for random intercepts because we expected that individuals would vary on their person-mean level of the study variables (the momentary assessments of interactions and affect). In addition, we also explored potential random slopes for the momentary reports of interactions, which assumed that the effects of social interactions would differ across individuals. According to our preliminary analysis, the inclusion of these random slopes did not make a difference for the observed pattern of effects when compared with the models only including random intercepts, and the effects were not significant in most of the models including random slopes. In the next chapter, we only presented the original simpler models that allowed for a random intercept but not for a random slope. However, we acknowledge that it is important to explore whether the effect would differ between individuals, and therefore we provided Appendix III for showing these results from models allowing for random slopes.

Multilevel linear regression models (Jahng & Wood, 2017) were conducted using the xtmixed function in Stata 13.0 (Corporation, 2007). The specified models aimed to test three sets of effects. First, we were interested in the between-person effects of reporting more interactions on the average level of affective states (Hypothesis 1) and the between-person effects of having more satisfactory interactions on the average level of affective states (Hypothesis 2). Second, it tested within-person effects that relative to one's average levels, whether the occurrence of social interactions was associated with affective states at the same and the subsequent occasion (Hypothesis 3 and 5); and also tested within-person effects that whether the satisfaction with social interactions was associated with the concurrent and subsequent level of affective states (Hypothesis 4 and 6). Third, it tested within-person effects that relative to one's average levels, whether the level of positive and negative affective states was associated with the subsequent level of satisfaction with social interactions (Hypothesis 7 and 8).

To separate the between-person and within-person effects, both the between- and within-person versions of each independent variable (IV) were included for each outcome in separate models. As our major purpose was to examine the within-person effects of independent variables on the outcomes separately from the between-person effect, the main independent variables were person-mean centered (observed value at one assessment occasion - person-mean). All analyses controlled for time-variant variables (i.e., location, physical level of the current activity), and other baseline participant characteristics (i.e., gender, age, marital status, education, BMI, retirement states, income, living arrangement, and absence of mobility limitations). The covariates measured at Level 2 (the person-level) were were grand-mean centered (person value – sample mean).

Chapter 5 Results

5.1 Sample Description

Table 5.1 presents the baseline characteristics of the entire sample (n = 78) as well as by sampling selection method. The mean age of participant was 58.97 (range = 50–70) years. Overall, 51.3% of participants were female, 87.2% were married, 60.3% received high school education or above, 79.5% reported being partially or fully retired, 62.8 % reported a monthly income level < RMB 3,000 yuan. In terms of living arrangement, 30.8% resided with both their spouse and adult children, 46.2% resided only with their spouse, while 19.2% reported living with other people.

Regarding the physical functioning, the average number of mobility limitations was 0.90 and 42.3% reported having at least one mobility limitation. Participants on average scored 28.31 for the Montreal Cognitive Assessment (MOCA) scale, indicating an overall satisfactory level of cognitive functioning. In terms of mental health, participants on average scored 2.22 on the geriatric depression (GDS) scale, suggesting that the sample was basically free from depressive symptoms. In sum, the study sample is considered relative healthy at baseline as evidenced by a few numbers of physical limitations reported, absence of cognitive impairment, and low scores in depressive symptoms.

We also tested the differences in the baseline characteristics (the last column in Table 5.1) between the participants selected through quota sampling (n = 59) with participants selected through snowball sampling (n =19). Percentages were compared using $\chi 2$ tests and means were compared using independent sample t-tests when normally distributed (when significant deviations from normal distribution were detected, the alternative method, Mann-Whitney U test was used). The participants in the snowball group were more likely to be male and older, which was resulted from the purposeful recruitment procedure we conducted. In addition to age and gender differences, these two groups did not differ significantly in terms of other baseline measures.

5.2 The Feasibility and Validity of the EMA Protocol

5.2.1 Objective compliance rates

The responses rates of each assessment interval were projected for male and female (see Figure 5.1). In the entire sample, participants reported a total number of 3,822 observations (91.5% of all scheduled assessments). For both female and male participants, the highest response rate (female 96.96%, male 90.6%) was reported at the third assessment interval (between 12:00 -14:00), and the lowest response rate (female 77.6% male 69.2%) was reported at the last assessment interval (between 20:00 - 22:00) that was optional for completion. Overall, Female participants reported higher response rates at all of the assessment intervals than male participants. In terms of the completion time, over half of the completed assessments (56.3%) were responded within 20 minutes of being prompted by the EMA application. The average completion time per assessment was 3.47 minutes for female participants ad 3.12 minutes for male participants. Only 0.39% of all completed responses presented missing values on one or more variables.

Table 5.1 Sample Characteristics by Sampling Method

	Overall		Quota		Snowball		Sig-test
	N	%	N	%	N	%	(p value)
Gender							
Men	38	48.7	28	47.5	10	78.9	.001
Women	40	51.3	31	52.5	9	21.1	
Marital status							
Married	68	87.2	52	88.1	16	84.2	
Not married/others	10	12.8	7	11.9	3	15.8	.656
Education level							
Less than high school	31	39.7	22	37.3	9	47.4	.104
High school or more	47	60.2	37	33.9	10	47.4	
Working status							
Full-time employment	11	14.1	10	16.9	1	5.3	.293
Partially Retired	10	12.8	9	15.3	1	5.3	
Fully Retired	52	66.7	36	61.0	16	84.2	
Unemployed	5	6.4	4	6.8	1	5.3	
Income (monthly)							
Less than RMB 3,000	49	62.8	37	62.7	12	63.2	.384
More than RMB 3,000	29	30.8	22	28.8	7	36.8	
Living arrangement							
Living alone	3	3.8	3	5.1	0	.0	.617
With spouse	36	46.2	27	45.8	9	47.4	
With spouse &children	24	30.8	19	32.2	5	26.3	
Others	15	19.2	10	16.9	5	26.3	
	Mean	SD	Mean	SD	Mean	SD	Sig-test
Age	58.97	5.76	58.02	5.63	61.95	5.22	.009
Body mass index (kg/m^2)	22.91	2.40	22.92	2.47	22.86	2.20	.923
GDS scores	2.22	1.50	2.27	1.62	2.05	1.08	.584
MOCA scores	28.31	1.04	28.25	1.09	28.47	0.84	.425
Mobility limitations	0.90	1.52	0.86	1.57	1.00	1.37	.737

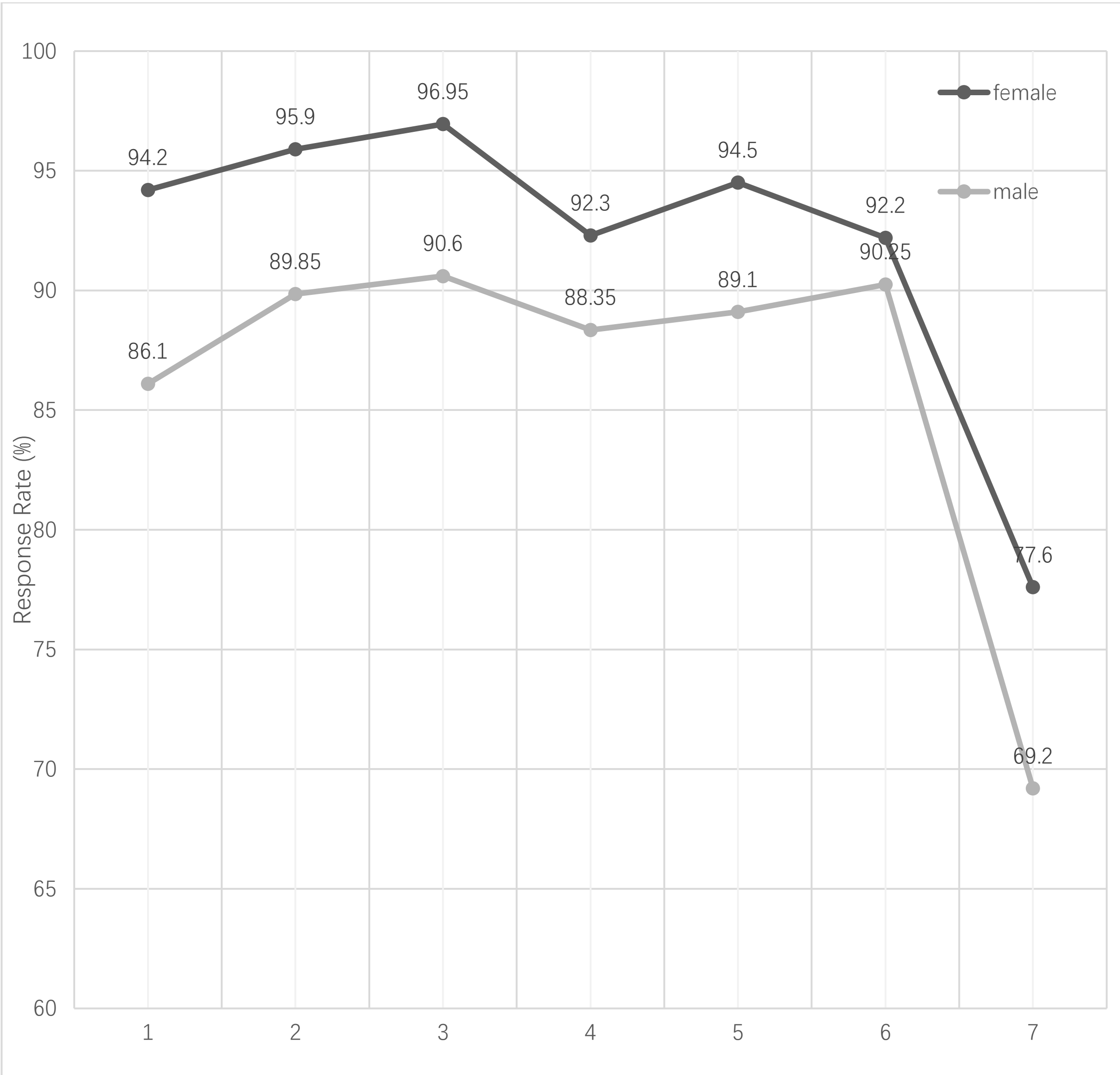

Figure 5.2 Average response rates at each assessment interval by gender

Notes: Assessment 1 = between 8:00-10:00; Assessment 2 = 10:00 – 12:00; Assessment 3 = 12:00 – 14:00; Assessment 4 = 14:00 – 16:00; Assessment 5 = 16:00 – 18:00; Assessment 6 = 18:00 – 20:00; Assessment 7 = 20:00 -22:00 (an optional assessment).

5.2.2 Subjective measures of feasibility and usability

Table 5.2.3 shows the subjective ratings on the feasibility issues by female and male participants separately. Female participants rated their experiences working on the EMA as relatively pleasant (mean 3.93, SD 1.12), challenging (mean 3.13, SD 1.03), and not boring (mean 1.23, SD 1.20). As for the assessments, the intensity of the prompts was perceived as relatively high (mean 3.01, SD 1.00), while the study duration and the length for completing one assessment were considered as reasonable. Participants perceived carrying the smartphones as inconvenient to some extent (mean 2.23, SD 1.01), somehow being interfered by the prompts with their daily activities (mean 2.01, SD 1.11), while reported few difficulties in working with the EMA procedures (mean = 1.02 and 0.98). Most female participants showed positive attitudes in terms of future participation (mean 4.07, SD 1.10) and recommending to others (mean 4.29, SD 0.98).

For male participants, the overall experiences working on the EMA were perceived as less pleasant (mean 2.45, SD 1.14) and more boring (mean 2.77, SD 1.27) than that were reported by female participants. Male participants did not differ from female participants in their ratings about the extent to which the protocol was perceived as challenging (mean 3.12, SD 1.09) or stressful (mean 2.14, SD 1.15). As for the assessments, male participants cosndiered the intensity of assessment as relatively high (mean 3.42, SD 1.02), and the time spent on completing one assessment as relatively long. They did not differ from female participants in terms of their rating about the one-week duration of the study. As for the interferences and difficulties encountered during the study, male participants expressed similar perceptions as their female counterparts, in that carrying the smartphones was somewhat inconvenient (mean 2.09 SD 1.21), they were somehow interfered by the prompts with their daily activities (mean 2.07, SD 1.14), while reported few difficulties in working with the EMA procedures (mean = 1.04 and 0.81). Regarding their willingness to participate in similar studies and recommend to others in the future, male participants gave positive answers, which were slightly lower than that of their female counterparts (mean 3.97 for future participation and mean 4.17 for recommendation).

Table 5.2.2 Self-rated protocol feasibility of female participants (n = 40)

Questions	Female (n = 40)	
	Mean	SD
Part 1. Overall experiences		
Do you consider the overall experience as pleasant?	3.93	1.12
Do you consider the overall experience as challenging?	3.13	1.03
Do you consider the overall experience as stressful?	2.12	1.11
Do you consider the overall experience as boring?	1.23	1.20
Part 2. The Procedures		
Completing one brief survey takes too much time.	1.98	0.86
It was too frequent to be prompted.	3.01	1.00
I think the one-week data collection period was too long.	2.11	1.20
Part 3. Difficulties and Burden		
The schedule heavily interfered with my daily life.	2.01	1.11
Having to carry the device with me is inconvenient.	2.23	1.01
It was not easy/direct enough to understand the questions.	1.02	0.45
Filling questions on the screen was not easy.	0.98	0.53

Part 4. Commitment		
I would involve in similar studies in the future.	4.07	1.10
I would recommend others for similar studies.	4.29	0.98

Questions	Male (n =38)	
	Mean	SD
Part 1. Overall experiences		
Do you consider the overall experience as pleasant?	2.45	1.14
Do you consider the overall experience as challenging?	3.12	1.09
Do you consider the overall experience as stressful?	2.14	1.15
Do you consider the overall experience as boring?	2.77	1.21
Part 2. The Procedures		
Completing one brief survey takes too much time.	2.04	0.88
It was too frequent to be prompted.	3.42	1.02
I think the one-week data collection period was too long.	2.09	1.21
Part 3. Difficulties and Burden		
The schedule heavily interfered with my daily life.	2.07	1.14
Having to carry the device with me is inconvenient.	2.09	1.06
It was not easy/direct enough to understand the questions.	0.81	0.51
Filling questions on the screen was not easy.	1.04	0.57
Part 4. Commitment		
I would involve in similar studies in the future.	3.97	1.13
I would recommend others for similar studies.	4.17	1.02

Note: This evaluation form originally included 14 questions, of which one question was not presented here because it was for other research purpose of the parent study.

5.3 The concurrent relationships between social interactions and momentary affect

5.3.1 Correlations and Descriptive Statistics for Main Study Variables

Preliminary analyses were performed for the purpose of describing the inter-individual and intra-individual characteristics of social interactions and affective experiences. Table 5.3.1 presented a summary of the descriptive statistics for the primary study variables and their correlations. Those who reported more occurrence of social interactions on average tended to report higher level of satisfaction with interactions, but such association was not significant at the within-person level, indicating that the occurrence of a given social interaction could be appraised as either satisfying or not within the same person across occasions. Regarding the between-person relationships of social interactions and affective states, those who reported more frequent social interactions on average also experienced more high-arousal positive affect, and less low-arousal positive affect and less low-arousal negative affect; while those who reported higher satisfaction with interactions tended to report more positive affect and less negative affect, regardless of the arousal. These associations have been replicated at the within-person level.

Also, we fitted a null (unconditional) model for each outcome variable of interest to calculate the intraclass correlations (reported at the bottom of Table 5.3.1). These models indicate that there was considerable within-person variance in satisfaction with interaction (ICC = 0.47); in high-arousal positive affect (ICC = 0.50), and in low-arousal affect (ICC = 0.45), and greater in high-arousal negative affect (ICC = 0.22) and in low-arousal positive affect (ICC = 0.32).

Table 5. 3. 1 Correlations and Descriptive Statistics for Main Study Variables

Variables (between-person)		1	2	3	4	5	6
1	Mean_ Occurrence	1	--	--	--	--	--
2	Mean_ Satisfaction	.233	1	--	--	--	--
3	Mean_ HPA	.276	.261**	1	--	--	--
4	Mean_ LPA	-.06	.185**	.415**	1	--	--
5	Mean_ HNA	-.03	-.061**	.182**	.180**	1	
6	Mean_ LNA	-.04	-.094**	.145**	.185**	.802**	1
Variables (within-person)		1	2	3	4	5	6
1	Momentary Occurrence	1	--	--	--	--	--
2	Momentary Satisfaction	.019	1	--	--	--	--
3	Momentary HPA	.169	.154**	1	--	--	--
4	Momentary LPA	-.16	.083**	.207**	1	--	--
5	Momentary HNA	.007	-.171**	.015	.009	1	--
6	Momentary LNA	-.10	-.137**	.008	.106**	.365**	1
	Variable Statistics						
Mean		0.61	7.49	4.11	4.01	0.23	0.60
Standard deviation		0.49	1.88	1.83	1.74	0.67	1.00
Range		0-1	0-10	0-10	0-8	0-8	0-8
Intraclass correlation (ICC)		--	0.476	0.507	0.449	0.223	0.317

Notes: HPA = high-arousal positive affect; LPA = low-arousal positive affect; HNA = high-arousal negative affect; LNA = low-arousal negative affect. *p < .05, **p < .01, ***p < .001.

1. Mean_Occurrence is the person-mean of social interactions representing the proportion of all occasions in which an interaction occurred for each person.

Momentary Occurrence is a dichotomous variable indicating whether the participant was having an interaction or not.

2. Mean_Satisafaction is the person-mean of the ratings about the level of satisfaction with all interactions. Momentary satisfaction is a continuous variable indicating the momentary satisfaction with interaction at each measurement occasion (scaled from 0 to 10, with higher score indicating higher satisfaction).
3. Mean_HPA is the person-mean of all the completed rartings on the level of high-arousal positive affect (scaled from 0 to 10, with higher score indicating higher level); Momentary HPA is the momentary value the participant reported on the level of high-arousal positive affect (scaled from 0 to 10). Mean_LPA and momentary LPA, Mean_HNA and momentary HNA and Mean_LNA and momentary LNA were measured following the same logics.

5.3.2 Occurrence of social interaction and momentary affect

To address Hypothesis 1 and 3, we tested a series of models examining the effect of the occurrence of social interaction (OSI) on affective states, including high-arousal positive affect (HPA), low-arousal positive affect (LPA), high-arousal negative affect (HNA), low-arousal negative affect (LNA). Both the between-person effects (i.e., comparing persons who have more interactions with persons who have less) and the within-person effects (i.e., comparing observations when a person was engaging in an interaction to observations when that person did not engage in an interaction) were examined in the same model (see Table 5.3.2). All models controlled for time-variant variables (i.e., location, physical level of the current activity), and other baseline participant characteristics (i.e., gender, age, marital status, education, BMI, retirement states, income, living arrangement, and absence of mobility limitations). Only results concerning fixed effects were discussed due to space concerns, while additional model information was provided in each table.

Table 5.3.2. Multilevel models examining the between-person and within-person effects of occurrence of social interactions (OSI) on momentary affect (n = 3,495)

Variables	HPA		LPA		HNA		LNA	
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE
OSI (within-person)	.36**	.05	-.57**	.05	.02	.02	-.24**	.03
OSI (between-person)	1.89*	.68	-.21	.67	-.08	.17	-.04	.32
Location (1 = home)	-.24**	.04	.25**	.04	-.02	.02	-.03	.03
No Physical Activity	-.13**	.04	.17**	.04	-.04	.02	.04	.03
Older Age	.36*	.35	-.21	.34	-.07	.09	-.34*	.16
Male	.59	.34	.21	.33	.07	.08	.22	.15
Married	.58	.42	.23	.40	.16	.10	.01	.19
Education	.25	.34	-.00	.33	-.05	.08	-.06	.15
Retired	.54	.43	-.12	.42	.14	.11	.41*	.20
Living with spouse	.32	.29	-.39	.28	.12	.07	.04	.13
Low Income	-.08	.33	-.10	.32	.04	.08	-.10	.15
No mobility	-.27	.29	-.02	.28	-.02	.07	-.22	.13
Intercept	1.63*	.77	4.03*	.75	.02	.20	.50*	.36
Variance Intercept	1.36		1.36		0.09		0.29	
Residual variance	1.66		1.66		0.35		0.67	

ICC	0.40	0.40	0.06	0.16
–2LL	5892.01	5892.	3244.51	4388.80

Notes: OSI (within-person) is a binary variable (0 = no interaction; 1= yes, having an inert action) measured for multiple times (up to 49 times) from the same individual for a week. OSI (between-person) is a continuous variable indicating the proportion of the occurrence of social interactions across all the scheduled assessments (up to 49 times) for each person. ICC = intraclass correlation. $*p < .05$, $**p < .01$, $***p < .001$. HPA = high-arousal positive affect; LPA = low-arousal positive affect; HNA = high-arousal negative affect; LNA = low-arousal negative affect.

Regarding the between-person relations (Hypothesis 1), those who had more interactions on average reported higher level of HPA than those who had fewer interactions. No between-person effects were observed for other affective states including LPA, HNA and LNA. In addition, older age was associated with experiencing more HPA and less LNA on average, while being retied was associated with more experiences of LNA on average.

As for the within-person relations (Hypothesis 3), when participants reported being engaged in social interactions, they reported higher level of HPA, lower level of LPA, and lower level of LNA than moments reporting no social interactions. We did not observe significant within-person covariation between social interactions and HNA. In addition, being at home (versus not being at home) and having no physical activity (versus having moderate or heavy physical activity) was related to higher level of LPA and lower level of HPA at the within-person level.

5.3.3 Subjective satisfaction with social interaction and momentary affect

To address Hypothesis 2 and 4, we tested a series of models examining the effect of the satisfaction with social interaction (SSI) on affective states. Both the between-person effects (i.e., comparing people who rated their interactions as more satisfactory to people who rated interactions as less satisfactory) and the within-person (i.e., comparing observations when an interaction was rated as more satisfactory to observations when that person rated an interaction as less satisfactory) effects were examined in the same model (see Table 5.3.3). All models controlled for the same covariates that were included in Table 5.3.2, and only results of the fixed effects were discussed.

Regarding the between-person relations (Hypothesis 2), those who rated their interactions as more satisfactory on average also reported higher level of HPA than those who rated interactions as less satisfactory. No between-person effects were observed for other affective states including LPA, HNA and LNA. In addition, older age and reporting no functional limitation were associated with lower level of LNA on average, while being retied was associated with higher level of LNA on average. As for the within-person relations (Hypothesis 4), when participants appraised an interaction as more satisfactory than that was typical for them (above their person mean level), they reported higher level of HPA and LPA, and lower level of HNA and LNA. In addition, being at home (versus not being at home) was related to higher level of LPA and lower level of HPA at the within-person level, and reporting no physical activity was related to lower level of LPA.

Table 5.3.3. Multilevel models examining the between-person and within-person effects of satisfaction with social interactions (SSI) on momentary affect (n = 2,739)

Variables	HPA		LPA		HNA		LNA	
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE
SSI (within-person)	.20***	.02	.08***	.01	-.08**	.01	-.09**	.01
SSI (between-person)	.24*	.09	.15	.09	-.01	.02	-.03	.04
Location (1 = home)	-.30***	.05	.32***	.05	-.01	.02	.02	.03
No Physical Activity	-.11*	.05	.05	.05	-.03	.02	.01	.03
Older Age	.34	.35	-.18	.32	-.07	.09	-.31*	.15
Male	.51	.33	.34	.30	.06	.08	.23	.14
Married	.63	.41	.14	.37	.17	.10	-.00	.18
Education	.13	.33	-.06	.30	-.05	.08	-.06	.14
Retired	.57	.42	-.14	.39	.15	.11	.39*	.19
Living with spouse	.30	.29	-.42	.26	.12	.07	.04	.12
Low Income	-.20	.32	-.15	.29	.04	.08	-.06	.14
No mobility	-.16	.27	-.09	.25	-.03	.07	-.24*	.12
Intercept	1.13*	.94	2.85**	.86	.07	.25	.64*	.42
Variance Intercept	1.29		1.07		0.08		0.24	
Residual variance	1.62		1.46		0.34		0.62	
ICC	0.39		0.35		0.05		0.13	
–2LL	4683.415		4534.900		2531.530		3353.041	

Notes: SSI (within-person) is a continuous variable indicating the momentary satisfaction with interaction at each measurement occasion and has been person-mean centered. SSI (between-person) is the person-mean of the ratings about the level of satisfaction with all interactions. *p < .05, **p < .01, ***p < .001. ICC = intraclass correlation. HPA = high-arousal positive affect; LPA = low-arousal positive affect;

HNA = high-arousal negative affect; LNA = low-arousal negative affect.

5.4 The lagged relationships between social interactions and momentary affect

5.4.1 Social interaction and the subsequent level of momentary affect

After the testing of the concurrent relationships, we moved to the examination of the time-lagged relationships between social interactions and momentary affect. To address Hypothesis 5 (the occurrence of social interaction was followed by higher level of positive affect and lower level of negative affect), we specified a series of models where the occurrence of social interaction (OSI) at prior time point (t-1) served as the independent variable for each affective outcome measured at the current time point (t). For each model, the between-person effect of the occurrence of social interaction was included, as well as the affective states measured at t-1 (for example, when the dependent variable was HPA at the current time point, we controlled for HPA at t-1). In addition, a set of covariates included in previous models were also controlled.

As shown in Table 5.4.1, we found significant within-person association between the occurrence of social interaction at t-1 with higher level of HPA and lower level of LPA. As the average time interval between two momentary reports was about 90 minutes, this result suggested that the within-person influence of being engaging in a social interaction on greater HPA and fewer LPA can last for 90 minutes. However, we did not find significant within-person association between the occurrence of social interaction at t-1 with LNA, indicating that even having an interaction was associated with fewer LNA, these effects became not significant after as much as 90 minutes.

To address Hypothesis 6 (higher satisfaction with social interaction was followed by higher level of positive affect and lower level of negative affect), we specified a series of models where the satisfaction with social interaction (SSI) at prior time point (t-1) served as the independent variable for each affective outcome. We did not present the results of these models here because no significant relationships were observed. These results indicated that even the satisfaction with an interaction was concurrently associated more positive affective states and less negative affective states (see the results reported in Table 5.3.3), these within-person effects may become minimal when individuals move towards next social interactions about 2 hours later.

Table 5.4.1. Multilevel models examining how occurrence of social interaction (OSI) at prior observation (t-1) influence momentary affect (n = 2,210)

Variables	HPA		LPA		HNA		LNA	
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE
OSI t-1 (within-person)	.13*	.06	-.17***	.05	-.04	.02	.06	.03
OSI (between-person)	1.63*	.72	-.13	.67	-.05	.18	-.03	.32
Affect at t-1	.111***	.02	.17***	.02	.12***	.02	.06**	.02
Location (1 = home)	-.25***	.06	.31***	.05	-.03	.02	.01	.03
No Physical Activity	-.15*	.06	.20**	.06	-.03	.02	.03	.04
Older Age	.42	.37	-.24	.35	-.10	.09	-.38*	.16
Male	.52	.36	.22	.33	.11	.09	.22	.16
Married	.64	.44	.18	.41	.17	.11	.00	.19
Education	.29	.35	-.05	.33	-.06	.09	-.09	.15
Retired	.46	.45	-.13	.42	.15	.12	.42*	.20
Living with spouse only	.33	.31	-.36	.28	.11	.08	.03	.13

Lower Income	-.12	.34	.01	.32	.03	.09	-.09	.15
No mobility	-.24	.30	.01	.28	-.02	.07	-.24	.13
Intercept	1.78*	.81	3.95***	.75	-.00	.21	.56	.36
Variance Intercept	1.47		1.28		0.09		0.28	
Residual variance	1.65		1.55		0.33		0.68	
ICC	0.44		0.41		0.06		0.17	
–2LL	3817.707		3749.278		2026.119		2820.589	

Notes: OSI (within-person) is a binary variable (0 = no interaction; 1= yes, having an inert action) measured for multiple times (up to 49 times) from the same individual for a week. OSI (between-person) is a continuous variable indicating the proportion of the occurrence of social interactions across all the scheduled assessments (up to 49 times) for each person. Affect at t-1 refers to HPA at t-1, LPA at t-1, HNA at t-1, LNA at t-1 respectively throughout the model 1 to model 4. HPA = high-arousal positive affect; LPA = low-arousal positive affect; HNA = high-arousal negative affect; LNA = low-arousal negative affect. ICC = intraclass correlation.

5.4.2 Positive affect and the subsequent level of satisfaction with social interaction (SSI)

To test Hypothesis 7a and 7b (higher level of positive affect was followed by higher satisfaction with future social interaction), we specified two models to examine the effects of the level of HPA and LPA at prior occasion (t-1) on the level of satisfaction with social interaction (SSI) at current occasion respectively. For each model, we controlled for the SSI measured at t-1, and the between-person variables of HPA and LPA. In addition, a set of covariates included in previous models were also controlled. As shown in Table 5.4.2, we did not find any significant relations between the level of HPA or LPA at t-1 with the subsequent level of SSI, thus being unable to find evidence supporting Hypothesis 7. Given that the lag intervals between two measurement of SSI tended to be around 123.5 (SD = 12.8) minutes apart, this result suggested that the effect of PA on subjective evaluation of social interactions did not preserve until the social occasions 2 hours later.

Table 5.4.2. Multilevel models examining how momentary positive affect at prior observation (t-1) influence subsequent level of satisfaction with social interaction (SSI) (n = 1,404)

Independent Variables	Coef.	SE	Independent Variables	Coef.	SE
HPA t-1 (within-person)	-.05	.02	LPA t-1 (within-person)	.02	.02
HPA (between-person)	.25*	.11	LPA (between-person)	.14	.12
SSI at t-1	.02	.02	SSI at t-1	.02	.02
Location (1 = home)	.17*	.08	Location (1 = home)	.18*	.08
No Physical Activity	.16*	.08	No Physical Activity	.15	.08
Older Age	.26	.36	Older Age	.37	.37
Gender	-.25	.35	Gender	-.18	.35
Married	-.08	.43	Married	.06	.43
Lower Education	.25	.34	Lower Education	.31	.35
Retired	-.07	.45	Retired	.07	.45
Living with spouse only	-.11	.30	Living with spouse only	.00	.31
Lower Income	.30	.33	Lower Income	.30	.34
No mobility	.32	.28	No mobility	.31	.29
Intercept	6.03***	.76	Intercept	6.07***	.89

Variance Intercept	1.30	Variance Intercept	1.35
Residual variance	1.87	Residual variance	1.87
ICC	0.33	ICC	0.34
–2LL	2529.50	–2LL	2532.17

Notes: HPA (between-person) and LPA (between-person) is the person-mean of the level of HPA and LPA respectively. SSI at t-1 refers to the satisfaction with social interaction at prior occasion. HPA at t-1 and LPA at t-1 refer to momentary reports of HPA and LPA at prior measurement occasion respectively. HPA = high-arousal positive affect; LPA = low-arousal positive affect. ICC = intraclass correlation.

5.4.3 Negative affect and the subsequent level of satisfaction with social interaction (SSI)

To test Hypothesis 7c and 7d (higher level of negative affect was followed by lower satisfaction with social interaction), we specified two models to test the effects of the level of HNA and LNA at prior occasion (t-1) on the level of satisfaction with social interaction at current occasion respectively. In each model, we controlled for the satisfaction with social interaction measured at t-1, and the between-person variables of HNA and LNA. In addition, a set of covariates included in previous models were also controlled. As shown in Table 5.4.3, higher level of HNA and LNA at t-1 were both significantly associated with lower satisfaction with subsequent social interaction. In other words, when a person reported a higher level of negative affect that usual, he or she tended to rate their subsequent interactions as less satisfactory. These results supported Hypothesis 8, suggesting that negative affective experiences could exert an enduring negative effect (last for about 2 hours) on individuals' future appraisal of social interactions.

Table 5.4.3 Multilevel models examining how momentary negative affect at prior observation (t-1) influence subsequent level of satisfaction with social interaction (SSI) (n = 1,404)

Independent Variables	Coef.	SE	Independent Variables	Coef.	SE
HNA t-1 (within-person)	-.14*	.06	LNA t-1 (within-person)	-.11*	.04
HNA (between-person)	-.34	.44	LNA (between-person)	-.12	.25
SSI at t-1	.02	.02	SSI at t-1	.02	.02
Location (1 = home)	.18*	.08	Location (1 = home)	.19*	.08
No Physical Activity	.15	.08	No Physical Activity	.14	.08
Older Age	.31	.37	Older Age	.29	.38
Gender	-.10	.35	Gender	-.08	.36
Married	.14	.44	Married	.09	.43
Lower Education	.29	.35	Lower Education	.31	.35
Retired	.13	.46	Retired	.13	.46
Living with spouse only	-.03	.31	Living with spouse only	-.06	.30
Lower Income	.27	.34	Lower Income	.24	.34
No mobility	.29	.29	No mobility	.27	.29
Intercept	6.64***	.71	Intercept	6.72***	.72
Variance Intercept	1.35		Variance Intercept	1.36	
Residual variance	1.87		Residual variance	1.87	
ICC	0.34		ICC	0.36	
–2LL	2530.27		–2LL	2529.86	

Notes: HNA (between-person) and LNA (between-person) is the person-mean of the level of HNA and LNA respectively. HNA = high-arousal negative affect; LNA = low-arousal negative affect. ICC = intraclass correlation.

5.5 Summary of results

In the basis of the results presented above, we provide a brief summary about whether the proposed hypotheses were supported or not.

Hypothesis 1

H1a: Individuals who reported more social interactions on average would have higher level of high-arousal positive affect (HPA) than those who reported less interactions. **(supported)**

H1b: Individuals who reported more social interactions on average would have higher level of low-arousal positive affect (LPA) than those who reported less interactions. (not supported)

H1c: Individuals who reported more social interactions would have lower level of high-arousal negative affect (HNA) than those who reported less interactions. (not supported)

H1d: Individuals who reported more social interactions would have lower level of low-arousal negative affect (LNA) than those who reported less interactions. (not supported)

Hypothesis 2

H2a: Individuals who reported more satisfactory social interactions on average would have higher level of HPA than those who reported less satisfactory interactions. **(supported)**

H2b: Individuals who reported more satisfactory social interactions on average would have higher level of LPA than those who reported less satisfactory interactions. (not supported)

H2c: Individuals who reported more satisfactory social interactions on average would have lower level of HNA than those who reported less satisfactory interactions. (not supported)

H2d: Individuals who reported more satisfactory social interactions on average would have lower level of LNA than those who reported less satisfactory interactions. (not supported)

Hypothesis 3

H3a: When individuals engaged in interactions, they reported greater HPA than situations when they did not engage in interactions **(supported)**

H3a: When individuals engaged in an interaction, they reported higher level of LPA than situations when they did not engage in interactions. **(opposite)**

H3a: When individuals engaged in an interaction, they reported lower level of HNA than situations when they did not engage in interactions. (not supported)

H3a: When individuals engaged in an interaction, they reported lower level of LNA than situations when they did not engage in interactions. **(supported)**

Hypothesis 4

H4a: When individuals appraised a social interaction as more satisfactory than that is typical for them, they reported higher level of HPA. **(supported)**

H4b: When individuals appraised a social interaction as more satisfactory than that is typical for them, they reported higher level of LPA. **(supported)**

H4c: When individuals appraised a social interaction as more satisfactory than that is typical for them, they reported lower level of HNA. **(supported)**

H4d: When individuals appraised a social interaction as more satisfactory than that is typical for them, they reported lower level of LNA. **(supported)**

Hypothesis 5

H5a: The occurrence of social interaction was followed by higher level of HPA. **(supported)**

H5b: The occurrence of social interaction was followed by higher level of LPA. **(opposite)**

H5c: The occurrence of social interaction was followed by lower level of HNA. (not supported)

H5d: The occurrence of social interaction was followed by lower level of LNA. (not supported)

Hypothesis 6

H6a: Higher interaction satisfaction was followed by higher level of HPA. (not supported)

H6b: Higher interaction satisfaction was followed by higher level of LPA. (not supported)

H6c: Higher interaction satisfaction was followed by lower level of HNA. (not supported)

H6d: Higher interaction satisfaction was followed by lower level of LNA. (not supported)

Hypothesis 7

H7a: Higher level of HPA was followed by higher interaction satisfaction (not supported)

H7b: Higher level of LPA was followed by higher interaction satisfaction. (not supported)

H7c: Higher level of HNA was followed by lower interaction satisfaction. **(supported)**

H7d: Higher level of LNA was followed by lower interaction satisfaction. **(supported)**

Chapter 6 Discussion

In this chapter, we first summarized the key findings of the present research and gave a comparison of the findings generated form our study with empirical findings of prior studies from relevant research fields (e.g., social gerontology, social psychology). Second, we provided a brief overview of the theoretical, methodological and practical contributions of the research findings for studying social and affective experiences of older population in urban China. Third, we discussed the limitations of the present research and offered recommendations for future researches.

6.1 The feasibility and usability of mobile-based EMA among older Chinese

Our study takes the first step to evaluate the feasibility and acceptability of mobile-based EMA for obtaining repeated measurement of social and affective experiences from community-dwelling older Chinese. First, the compliance rate reported by our sample was 91.5%, which was higher than that have been reported by most prior research (between 55% – 87%) across different study populations (Cain et al., 2009). While this high compliance rate might have been attributed to a variety of factors (e.g., most of the participants were healthy and highly independent), our experiences suggested the great promise of user-based elements for the enhancement of participants' adherence to the protocol (Liu & Lou, 2018). Specifically, we incorporated three key user-based elements into the original EMA protocols. The first element was adding an optional time interval without sounded alarm, which has been useful for accommodating individuals' different waking patterns. The second element was allowing participants to start data entry either before or after being prompted. We added this element because during the pilot study, many participants expressed that they felt stressful when they needed to response timely without a delay more than 20 minutes and would prefer more autonomy in deciding the response time. Th third element was the provision of self-monitoring by displaying the number of completed and uncompleted assessments in the welcome page. We believed that these user-based elements are unique and significantly enhanced participant autonomy, comfort and satisfaction (Shiffman et al., 2008), making our EMA design distinct from most of prior EMA protocols. Of note, the optional assessment (without alert) scheduled between 8:00-10:00 pm also received a response rate above 73%, underscoring the potential of self-monitoring coupled with system alert for the improvement of users' adherence to sophisticated EMA procedures (Brose & Ebner-Priemer, 2015).

Although the response rates reported in our sample were relatively high, we also identified some variations across multiple gender-age groups. Most importantly, the current protocol seems to be more feasible for women, because in our sample, female participants not only reported higher compliance level but also demonstrated a higher level of interest and satisfaction than male participants with working on this protocol. Pervious EMA studies tended to use samples that were predominated by female participants (Ramsey et al., 2016), thus providing little information concerning the potential gender gap in compliance among older population (Courvoisier, Eid, & Lischetzke, 2012). Here we speculated that the main topic of the study (social and affective experiences) might be more interesting for women and they may also feel more comfortable than men with self-discourse during daily engagement (R. W. Larson, Richards, & Perry-Jenkins, 1994; Scollon, Prieto, & Diener, 2009). In future research, more comprehensive strategies should be considered to enhance older men's adherence to and involvement in the study(Fritz et al., 2017).

Within the male participants, it was somewhat unexpectedly that the response rate of the younger group (aged between 50-59) was lower than that of the older group (aged between 60-69). One possible explanation could be the differences in time availability. The older group that was comprised of male retirees whereas most of the male participants in the younger group were still in the labor market and may have more difficulties in adhering to the protocol because of their job demands. Importantly, these male workers in our sample also showed a lower level of involvement in the early preparing stage of the data collection such as the training session and warm-up days. In contrast, their older counterparts tended to spend more time on the training session and supervised use of the application, and once encountering problems during the practices, they tended to ask for and receive technical assistance more rapidly. This finding delivers an important message, that is, older age itself may have minimal impact on the adherence, but the adequate training and practice as well as timely assistance matter for the maximization of compliance (Kuntsche & Labhart, 2013; Ramsey et al., 2016).

6.2 The between-person influence of social interactions on high-arousal positive affect (HPA)

To date, most of the evidence connecting features of social relationships to affective experiences has been generated from studies focusing on between-person differences. In general, these evidences tended to support the view that individuals who are more frequently engaged in social interactions, especially those positive social interactions (e.g., perceived as pleasant and satisfactory) also experience better affective outcomes in terms of more frequent positive affect and less frequent negative affect (Liu et al., 2018; Vogel et al., 2017). Our study significantly contributed to this literature by disaggregating the between-person effects of social interactions on affective experiences from the within-person effects, moving towards to a more precise understanding of the role social interactions played in shaping long-term individual differences in affective experiences (Bernstein et al., 2017; Papp, 2004; Snippe et al., 2016).

According to our results, both the quantitative and qualitative aspects of social interactions matter for the between-person differences in high-arousal positive affect (HPA). Regarding the quantitative aspects of social interactions, individuals who had more frequent social interactions experienced higher level of HPA than those who had less frequent social interactions. This result was consistent with prior research indicating the affective benefits of being with social partners rather than being alone during later life (Hawkley, Burleson, Berntson, & Cacioppo, 2003; von Humboldt et al., 2015). Similarity, we found that individuals who perceived their interactions as more satisfactory also experienced higher level of HPA than those who perceived their interactions as less satisfactory. This result provided new evidences to the emerging body of studies that suggested the importance of the quality of relationship for shaping individual differences in subjective well-being (Rafaeli, Cranford, Green, Shrout, & Bolger, 2008; Rook, 2001).

On the other side, it seems that the between-person effects of social interactions were not significant on either high-arousal or low-arousal negative affective sates.

Keeping with the two-factor theory of affect that regarded positive affect (PA) and negative affect (NA) as two independent factors (Posner et al., 2005; Watson & Tellegen, 1985), this result added new supporting evidence to the underlying hypothesis: the intra-individual differences in PA were influenced more by the environment while the intra-individual differences in NA were more genetically driven and influenced more by personality traits (Röcke & Brose, 2013; Yao, Robert, & Sophie von, 2016). While the existing evidences concerning this hypothesis were primarily generated from highly controlled experimental settings, our results documented the predicted effects of naturally occurring social interactions, suggesting that these relationships are generalizable to real life contexts, and provide valuable insights into the inference about the mechanism of how social interaction in everyday life leads to long-term affective well-being outcomes (Riediger et al., 2014; Wilhelm & Schoebi, 2007).

6.3 Linking aspects of social interactions to the within-person variations in affect

The core value of EMA-based research for connecting the external environment to people's internal state is the within-person approach, which examines how social interactions are associated with short-term changes in affective states of the same person across moments, hours and events. Adopting this within-person approach, the present study was able to link aspects of social interactions to the intra-individual variations in affective responses (Vogel et al., 2017). Consistent with the view that being socially connected benefitted affective well-being (Y. Chan & Lee, 2006; Huxhold et al., 2014), our findings successfully linked the occurrence of as well as the satisfaction with a given social interaction with more favorable affective responses. Specifically, when individuals were engaged in social interactions, they reported greater HPA and fewer LNA than moments when they were not. As expected, the satisfaction with interaction was an important predictor of concurrent level of affect. When participants appraised an interaction as more satisfactory than that was typical for them, they reported higher level of HPA and LPA, and lower level of HNA and LNA. By revealing these associations, our research findings are complementary to the long-term influence of social relationships on global well-being that were found on a between-person basis, and offered new insights into the covariation of social contact and affective responses on a within-person basis (Timmermans et al., 2010).

Beyond the examination of purely concurrent associations, our study also investigated the time-lagged association of social interactions and affective responses, in specific, whether aspects of social interactions measured at t-1 influenced affective responses at time t. We found a significant lagged association, in the expected direction, between the occurrence of social interaction at t-1 with the level of HPA. It suggested that the within-person effects of having a social interaction on greater HPA are not limited to concurrent effects but extend to affective responses at next measurement point (about an average of 90 minutes later). Unexpectedly, the lagged association between the occurrence of social interaction at t-1 with fewer LPA was also significant, suggesting that the inhibiting effect of having social interactions on LPA may last for at least 2 hours. Indeed, engaging in social interactions can be a double-edged sword by exhibiting significant enduring arousal-specific effects on positive affect, that is, having social interactions enhanced HPA but inhibited LPA. This finding is contradictory to the long-held view that a lack of social interaction is harmful, providing one of the first evidence of the affective benefits people enjoy during momentary solitude (Kira S. Birditt et al., 2018; Jennifer Christina Lay, 2018; Jennifer C Lay et al., 2018; McIntyre, Watson, Clark, & Cross, 1991).

In addition, we did not find evidence supporting the hypothesized lagged associations between the satisfaction with social interaction and subsequent affective states, we did not find any significant effects, even though there were evidences of strong associations between these measurements at the concurrent analysis. These null findings may suggest that the effects of one's subjective evaluation of an interaction on affective responses were not as sustainable as what we expected (cannot persevere until social interactions that occurred to people 2 hours later). Alternatively, it may be attributable to the way we measured participants' appraisal of an ongoing social interaction – asking our participants "to what extent you felt satisfied with an ongoing interaction" using a 10-point scale. It is possible that this measure was only able to capture the positivity of an interaction that was assumed to have an immediate rather than an enduring impact on affective responses. Future work should be devoted to the development of more explicit measures to capture qualitative aspects of social interactions that would be important for clarifying the time-varying effects of social experiences on affective states (Newsom, Rook, Nishishiba, Sorkin, & Mahan, 2005).

6.4 Negative affect predicted future satisfaction with social interactions

Prior experimental research has shown a long-standing interest in testing the reciprocal causal effects relating qualitative features of social event and affect.

Many of these studies have linked negative features of social interactions (e.g., poor communication quality, arguments) to subsequent negative affect such as distress and anxiety (McIntyre et al., 1991), but few of them provided evidences supporting the reverse direction of this relationship, that is, negative affect predicted future interaction quality (McIntyre et al., 1991). Thus, our study showed a particular interest in examining how affective states experienced at prior time point might influence subsequent evaluation of social interaction (specified as Hypothesis 4d), which was believed to offer valuable insights into the intra-individual affective process within the dynamic influence of social environment (Liu et al., 2018; Timmermans et al., 2010).

According to our lagged analyses, both HNA and LNA were associated with lower satisfaction with subsequent social interaction, while these lagged relations were not significant for HPA or LPA. Given that the lag intervals between two measurement of interaction satisfaction was over 120 minutes apart in our study, negative affective experiences (independent of the arousal dimension) could exert an enduring negative effect on individuals' future appraisal of social interactions that occurred two hours later. However, the effect of positive affective experiences on subjective evaluation of social interactions did not preserve until individuals move towards future interactions after a while. Consistent with the existing literature that was primarily based on experimental and diary studies, our findings support the view that negative affect have a more persistent impact on individual's cognition-related experiences than do positive affect (Kira S Birditt et al., 2018; Hepp et al., 2017; Martinez-Corts et al., 2015). Therefore, the potency of negative affect should lengthen the duration of their influence on individuals' subjective evaluation of the surrounding social environment. As our finding suggested that initially greater negativity in affective experiences fostered greater negativity in how people perceive their future interactions, it may serve as a starting point for subsequent studies to examine the causal structure of negative affect and individuals' perceptions about naturally occurring social experiences. Notably, while the current study was only able to collect a maximum of seven measurements of social interaction each day, future studies should consider the feasibility of using denser measurement assessment. By doing so, researchers could investigate more complex relationship between two or more variables of interest with better control of confounding factors (e.g., contextual factors) (Jahng, Wood, & Trull, 2008).

6.5 Contributions of the current study

6.5.1 Theoretical Contribution

This study provided theoretical implications for the dynamic interplay of social environment and affective experiences in older people's daily life. First, our conceptualization of social interactions was based on a multidimensional perspective that highlighted the importance of both quantitative and qualitative aspects of social interactions for shaping affective experiences. In line with this view, our findings successfully linked the occurrence of as well as the satisfaction with social interaction with more favorable affective responses on the within-person basis. These findings offered new insights into the covariation of social and affective experiences within the same person and were complementary to the well-established between-person influence of social contact on individual well-being.

Second, our study also examined the between-person effects of social interactions separately from the within-person effects. This is important because between-person and within-person approaches ask fundamentally different research questions, and therefore it cannot be assumed that the relationship observed from the between-person level could be in the same direction at within-person level. We found that at the between-person level, individuals who reported higher frequency of engaging in social interactions especially those satisfactory interactions also experience more favorable affective outcomes. By properly disaggregating within-person from between-person effects, our result precisely revealed the importance of the long-term properties of social interactions (e.g., the frequency and the average level of positivity) for shaping positive affective experiences of older adults.

Third, this study examined lagged relations of moment-to-moment social interactions with intra-individual variations in affective experiences. By doing so, our study provided interesting information concerning the temporal pattering in social relationships and affective experiences. Specifically, we provide the first evidence of the mutual inhibiting relations of having social interactions and low-arousal positive affect (LPA), which gave implications on the co-occurring pattern of solitude-seeking and positive internal states and enhanced our understanding of the multifaceted nature of momentary solitude experiences in later life (Jennifer Christina Lay, 2018). Regarding the reciprocal causal effects relating qualitative features of social event and affect, our study provided a new evidence that initially greater negativity in affective experiences fostered greater negativity in how people perceive their future interactions. In sum, these lagged relations provide valuable insights into the mechanistic inferences on aspects of social environment and affective well-being.

6.5.2 Methodological Contribution

This study developed a mobile-based EMA protocol to collect (real-time or near real-time) data of older people's day-to-day experiences of social interactions and concurrent affective states. It provided one of the first evidence concerning the design and implementation issues when using a mobile-based EMA protocol with older population, as well as the first evaluation of the full compliance patterns and subjective ratings of the feasibility and usability of this protocol in a sample of older adults living in urban China. By focusing on the methodological unknowns noted previously regarding the reliable use of sophisticated EMA, our study may inform future design and implementation strategies to enhance participant adherence with the mobile-based EMA protocols in older populations.

Regarding the design and implementation of mobile-based EMA protocol, we highlighted the incorporation of user-customized elements as key area for improvement in future development and applications of mobile-based EMA, which could enhance participant autonomy, comfort and adherence with the study (Shiffman et al., 2008). Also, we highlighted the importance of proper training, adequate practice and supervision, monitoring and timely assistance for enhancing compliance. In particular, we developed a four-step training procedure that has been proven to be useful for enhancing participants' confidence in working on the EMA protocol. In general, our study has shown that mobile-based EMA protocols can be both challenging and successful in collecting real-time data from older population. Future studies could learn from our experiences especially in terms of how to deal with detailing challenges faced during the implementation stage to improve utility and compliance.

6.5.3 Empirical Contribution

First, our research findings can be useful for clinicians to identify the key features of daily social interactions that may boost or deflates individuals' daily affect. In particular, our findings underscore the importance of satisfactory social interactions for enhancing favorable affective outcomes. This finding may inform policy makers and practitioners to deliver more programs to enhance older people's involvement in a variety of activities that provide older people opportunities for high-quality communication with others.

Second, our research finding can also inform the development of programs that aimed to promote healthier social environments for meaningful engagement and the betterment of affective well-being in old age. In specific, our results revealed the bidirectional effects of having social interactions and fewer low-arousal positive affect (LPA), adding new evidence to the multifaceted nature of solitude experiences in daily life (absence of interacting partner). Based on this finding, we suggested that clinical attention should be shifted from the 'more is better' paradigm (e.g., advocate more participation) to the development of programs that reduce the gap between the desired level and the actual level of social interactions, which would foster more positive affective experiences when older people were situated in different social contexts.

Furthermore, practitioners and researchers in the field of clinical settings may gain important insights from our study into the future development of momentary ecological interventions that incorporate mobile technology (such as self-monitoring technology and interactive technology) into psychosocial treatments. Of note, our finding suggested that initially negative affective experiences, which were often accompanied by negative evaluation of social interactions, reinforced greater negativity in subjective experiences of future interactions. This finding provided valuable information about the timing (critical periods) that proper interventions should be provided for older adults to minimize their negative experiences of the social environment. For example, informational support (e.g., reminders to reappraisal a negative social event) could be provided automatically when a reinforcing pattern of negative social and affective experiences was detected. Also, it is promising for practitioners to develop more personalized interactive interventions for the maximization of affective benefits drawing from individual's social dynamics, for instance, once the individual become aware of his or her changes in emotional patterns (e.g., increased negative affect after a negative social event), she or he could prompt an interactive feedback that help to reduce foreseeable negative outcomes.

Specific to the field of social work practice, our study made the following suggestions. First, more efforts should be directed to the promotion of meaningful engagement in social activities. Social workers can develop interventions that promote contextualized social interactions in community and LTC settings (enhance LPA during solitude activities such as meditation; foster HPA during community-based social activities). Second, subjective evaluation is especially important for influencing affective well-being. Social workers may need to encourage and educate people to be aware of their subjective experiences during different social situations. Third, negative affect can have a long-lasting impact on people's social experiences. Social workers provide interventions focusing on how to break the negative cycle when it occurred to people (bad contact was accompanied with negative affect, which may lead to bad contact in the future).

6.6 Limitations of the current study

6.6.1 Limitations of the sampling methods

First, this study was designed to collect the data in just one case city, which would unavoidably limit the generalizability of the results to the situation of urban China, where different regions and cities departs from each other in terms of both social and economic conditions. Second, the study would only be able to use a convenience sampling method to recruit a relatively small number of participants. By adopting this method, there would be some possible sample bias, for example, persons who were highly educated and more socially active would be more likely to engage in this study. Furthermore, our specificity of the inclusion criteria led to a sample of primarily healthy urban seniors, which may have limited the possible generalizability of our research findings to other older populations in mainland China (e.g., rural aging population; cognitively impaired elders). Despite of these, the recruitment of participants in the current study has tried hard to feasibly accomplish the research objectives. With more available resources, future research may consider the possibility of expanding the period of participant recruitment, as well as a combination of different methods for recruiting hard-to-reach participants.

6.6.2 Limitations of the design and implementation of the EMA protocol

The first limitation regarding the design of the EMA protocol is about the intensity of our scheduled assessment. Our study was only able to collect a maximum of seven reports on the targeted experiences per day, which was considered a medium level of intensity (the intensity of assessment reported across EMA studies in this field ranged from 3 to 15 reports per day). Although our decision on this intensity was primarily based on the concern about the acceptability and participant burden, we acknowledged that this intensity has largely limited the possibility of this study exploring temporal precedence of associations. Moreover, it is possible that the density of assessment in the current study was not enough for observing significant lagged relations between the variables we tested, for example, the lagged effect of satisfaction with social interaction on subsequent affective states was not significant. Future research may consider using a more intensive assessment design. With greater density and better control of contextual factors, researchers could investigate more complex relationship between two or more variables of interest and make stronger inferences about causality.

In addition to the issue about assessment intensity, there are several limitations relevant to the implementation and evaluation of the mobile-based EMA protocol. For example, malfunction issues occurred during the data collection period, which might have caused some participant burden. Also, we did not explore whether older people's attitudes towards and proficiency with mobile technology might influence their compliance, nor did we include several factors (such as interest in the study topic and motivation to participate in) that may contribute to their adherence in the data collection process.

6.6.3 Limitations of the selection and design of the EMA instruments

Regarding our selection and development of the EMA instruments, our study also encountered the challenge that all EMA studies faced, that is, the unavailability of standardized items/scales with proven psychometric properties. Specific to the assessment of aspects of social interaction, out study primarily focused on the occurrence of and the self-rated satisfaction with a given interaction. We believe that additional aspects of social interactions may be important for boosting or deflating specific affect in daily life (e.g., whether the occurrence of social interactions matched with individuals' desire for solitude). Also, it is possible that our measure of self-rated satisfaction with an interaction was only able to capture the positivity of an interaction that may be more likely to exert immediate rather than long-lasting impact on affect. Future work should be devoted to the development of more theory-driven and explicit measures to capture the mixture of qualitative aspects of social interactions (e.g., some interactions perceived as satisfactory but also demanded efforts or even stressful). While the current study only adopted the quantitative approach, future research may also consider incorporating qualitative approach to capture more intrinsic factors of older retirees' experiences when being socially engaged or not.

As for the assessment of affective states, we only used two or three items for each affect dimension to reduce participant burden. Although these items have been widely used s and have been proven to be sensitive enough for detecting short-term changes in affect with adequate psychometric properties in EMA studies (H. Chui et al., 2014; Vogel et al., 2017), it is possible that our results were influenced by the selection of specific affect items. Therefore, it is important that subsequent studies use a larger number of affect items to strengthen reliability and validity of the valence and arousal dimensions.

6.7 General conclusion

Social environment plays a central role in influencing affective experiences since the earliest days of human life. A substantial amount of researches indicated that individuals who are more frequently engaged in social interactions, especially those pleasant and satisfactory interactions, also experienced more favorable affective outcomes. Taking the advantage of EMA, we contribute to this field via three important ways: (1) extending prior research to consider momentary (both quantitative and qualitative) aspects of social interactions that occurred naturally to people in daily life settings; (2) separating the between-person and within-person influence of social interactions on affective experiences; and (3) taking the first step to testing the reciprocal relations of social interactions and momentary affective states.

First, our study adopted a multidimensional framework of social interactions that highlighted the importance of both quantitative and qualitative aspects of social interactions for shaping affective experiences. On the within-person basis, we linked the occurrence of and higher satisfaction with social interactions with more favorable affective responses, complementing to the well-established between-person influence of social contact on subjective well-being. Future studies can build on this conceptual framework and examine additional qualitative features of social interactions and their impacts on the affective well-being of older individuals.

Second, our study provided evidence of how the average level of social interactions influence individual differences in affective experiences. By properly disaggregating within-person from between-person effects, our results precisely revealed the importance of more frequent and satisfactory interactions for enhancing positive affective experiences of older adults. Future studies could extend from the current research to separate the between- and within-person influence of social interactions on a broader array of health and well-being outcomes.

Third, this study took the first step to examine the potentially causal relationships linking social interactions with affect. Specifically, our finding revealed the mutual inhibiting relations of having social interactions and low-arousal positive affect (LPA), as well as the fact that initially greater negativity in affective experiences fostered greater negativity in how people perceive their future interactions. These lagged relations provide valuable insights into the mechanistic inferences on aspects of social environment and affective well-being.

In summary, our study evidenced that social interaction is a powerful influence on affective well-being in daily life. In practice, our research findings can be useful for clinicians to identify the key features of day-to-day social interactions that influence individuals' affective outcomes, as well as inform the development of programs to that aimed to promote healthier social environments for older people's meaningful engagement and the betterment of affective well-being. Clinical attention should be shifted from the 'more is better' paradigm (e.g., advocate more participation) to the development of programs that focus on minimizing the gap between the desired versus actual level of social interactions, which would ultimately foster more positive affective experiences when older people were situated in different social contexts. Furthermore, practitioners (e.g., community mental health professionals, social works, consultant) may gain insights from our study into the future development of momentary ecological interventions that incorporate mobile technology into psychosocial treatments. In specific, proper interventions such as informational support (e.g., reminders to reappraisal a negative social event) and interactive feedback could be provided when a reinforcing pattern of negative social and affective experiences was detected, which hold great potential for the maximization of affective benefits drawing from individual's social dynamics.

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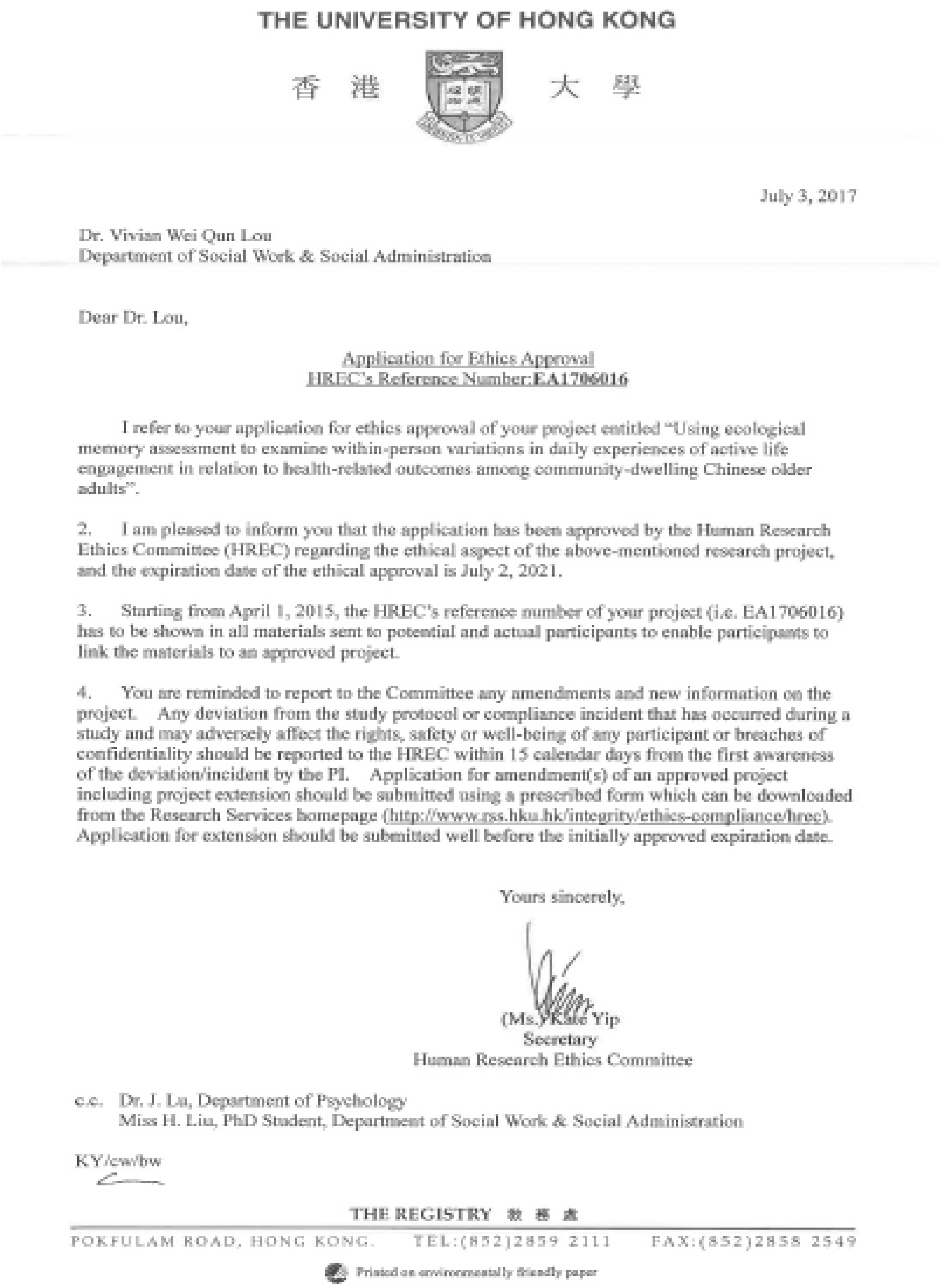
Appendices

Appendix I

Ethical approval



THE UNIVERSITY OF HONG KONG

香 港 大 學

July 3, 2017

Dr. Vivian Wei Qun Lou
Department of Social Work & Social Administration

Dear Dr. Lou,

Application for Ethics Approval
HREC's Reference Number:EA1706016

I refer to your application for ethics approval of your project entitled "Using ecological memory assessment to examine within-person variations in daily experiences of active life engagement in relation to health-related outcomes among community-dwelling Chinese older adults".

2. I am pleased to inform you that the application has been approved by the Human Research Ethics Committee (HREC) regarding the ethical aspect of the above-mentioned research project, and the expiration date of the ethical approval is July 2, 2021.

3. Starting from April 1, 2015, the HREC's reference number of your project (i.e. EA1706016) has to be shown in all materials sent to potential and actual participants to enable participants to link the materials to an approved project.

4. You are reminded to report to the Committee any amendments and new information on the project. Any deviation from the study protocol or compliance incident that has occurred during a study and may adversely affect the rights, safety or well-being of any participant or breaches of confidentiality should be reported to the HREC within 15 calendar days from the first awareness of the deviation/incident by the PI. Application for amendment(s) of an approved project including project extension should be submitted using a prescribed form which can be downloaded from the Research Services homepage (http://www.rss.hku.hk/integrity/ethics-compliance/hrec). Application for extension should be submitted well before the initially approved expiration date.

Yours sincerely,

(Ms.) Kate Yip
Secretary
Human Research Ethics Committee

c.c. Dr. J. Lu, Department of Psychology
Miss H. Liu, PhD Student, Department of Social Work & Social Administration

KY/cw/bw

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Attachment II

EMA Questionnaire for “Everyday Social and Affective Experiences Brief Survey’

Questions		Response options	
Part A Context	Q1. The primary activity you are doing now	● Eating	● Personal care
		● Household work	● Paid working
		● Caregiving	● Volunteering
		● Informal helping	● Reading/listening to the radio
		● Watching TV	● Using electronics
		● Leisure activities	● Exercises
		● Shopping	● Locomotion
		● Medical-related activities	● Communication
		● Learning activities	● Other activities
	Q2. Duration of performing the primary activity	● Between 5-10 minutes	● Between 11-20 minutes
		● Between 21-30 minutes	● Between 31-60 minutes
		● Over 60 minutes	
	Q3. Environment	● Home	● Indoor (outside home)
		● Outdoor	
	Q4. Current level of physical activity	● No physical activity	
		● Light physical activity	
		● Moderate physical activity	
		● Heavy physical activity	
Part B Features of Social	Q5. Who is with you? (Branch question) *Answer Pattern 1*		
	● Alone	Q6. Have you interacted with someone since last data entry?	

Interaction		● No interaction (skip Q7)	
		● Yes, less than 10 minutes ago	
		● Yes, about 10-30 minutes ago	
		● Yes, about 31-60 minutes ago	
		● Yes, over 60 minutes ago	
		Q7. Who did you interact with?	
		◆ Spouse	◆ Other families
		◆ Parent	◆ Friend/neighbor
		◆ Children	◆ Grandchildren
		◆ Co-worker	◆ Others
	Q5. Who is with you? (Branch question) *Answer Pattern 2.*		
	◆ Spouse	◆ Other families	
	◆ Parent	◆ Friend/neighbor	
	◆ Children	◆ Grandchildren	
	◆ Co-worker	◆ Others	
	Q8. Are you interacting now?		
	● No (back to answer Q6 and Q7)		
	● Yes		
	Q9. Duration of the ongoing/latest interaction you indicated before. For the participants who answered affirmative in Q6 or Q8, they were instructed to answer additional seven questions (Q9-Q15).		● Between 5-10 minutes
			● Between 11-20 minutes
			● Between 21-30 minutes
			● Between 31-60 minutes
			● Over 60 minutes
	Q10. Who initiated this interaction？		● Me

<table>
<tr><td rowspan="4"></td><td colspan="7" rowspan="2"></td><td colspan="3">● The partner</td></tr>
<tr><td colspan="3">● Hard to say</td></tr>
<tr><td colspan="10">Q11. Rate your satisfaction with this interaction from 0 to 9, with higher value indicating higher satisfaction.</td></tr>
<tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr>
<tr><td rowspan="20">Part C
Momentary
Affect</td><td colspan="10">Q17. The extent you feel quiet</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q18. The extent you feel happy</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q19. The extent you feel excited</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q20. The extent you feel sad</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q21. The extent you feel alert</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q22. The extent you feel anxious</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q23. The extent you feel sleepy</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q24. The extent you feel calm</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q25. The extent you feel irritated</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
<tr><td colspan="10">Q26. The extent you feel lonely</td></tr>
<tr><td colspan="2">not at all</td><td colspan="2">a little bit</td><td colspan="2">moderately</td><td colspan="2">quite a bit</td><td colspan="2">very much</td></tr>
</table>

Appendix III

Supplementary Tables. Estimations from multilevel models (allowing random slopes) examining the between-person and within-person effects of occurrence of social interaction (OSI) on affect

Variables	HPA		LPA		HNA		LNA	
	Coef.	SE	Coef.	SE	Coef.	SE	Coef.	SE
OSI (within-person)	.36**	.05	-.57**	.05	.02	.02	-.24**	.03
OSI (between-person)	1.90*	.68	-.21	.67	-.08	.17	-.04	.32
Location (1 = home)	-.24**	.04	.25**	.04	-.02	.02	-.03	.03
No Physical Activity	-.12**	.04	.17**	.04	-.04	.02	.04	.03
Older Age	.38*	.35	-.21	.34	-.07	.09	-.34*	.16
Male	.57	.34	.21	.33	.07	.08	.22	.15
Married	.54	.42	.23	.40	.16	.10	.01	.19
Education	.22	.34	-.00	.33	-.05	.08	-.06	.15
Retired	.54	.43	-.12	.42	.14	.11	.41*	.20
Living with spouse only	.36	.29	-.39	.28	.12	.07	.04	.13
Low Income	-.10	.33	-.10	.32	.04	.08	-.10	.15
No mobility	-.25	.29	-.02	.28	-.02	.07	-.22	.13
Intercept	1.66*	.77	4.11**	.75	.02	.20	.48*	.36
Random effects								
Variance Intercept	1.37		1.30		0.09		0.29	
Variance (OSI)	0.18		0.37		0.02		0.16	
Covariance (intercept,	-.10		-.25		-.00		-.11	

OSI)				
Residual variance	1.62	1.50	0.35	0.64
–2LL	5970.29	5849.71	3240.59	4346.89

Notes: OSI (within-person) is a binary variable (0 = no interaction; 1= yes, having an inert action) measured for multiple times (up to 49 times) from the same individual for a week. OSI (between-person) is a continuous variable indicating the proportion of the occurrence of social interactions across all the scheduled assessments (up to 49 times) for each person. ICC = intraclass correlation. *$p < .05$, **$p < .01$, ***$p < .001$. HPA = high-arousal positive affect; LPA = low-arousal positive affect; HNA = high-arousal negative affect; LNA = low-arousal negative affect.